Projector-Resolved Seed Calculus VII

Quantum Seed Calculus: Symmetry Packets, Circuits, Effective Dynamics, and Quantum Spacetime

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Abstract

This paper develops the finite quantum realization of the projector-resolved seed calculus. A quantum seed is propagated by a declared gate or observable grammar, resolved into symmetry packets, transferred through circuit paths, and reconstructed exactly. For finite observable algebras, the dimension of the orbit generated by a seed is computed from the ranks of its multiplicity matrices, yielding an explicit quantum seed-rank formula. Cyclic symmetry projectors and tensor/polyphase refinement are proved in a common operator language. Quantum circuits admit an exact path expansion over intermediate packet addresses, and unitarity becomes a packetwise conservation law. For Hamiltonian evolution, exact symmetry produces invariant packets, while a Duhamel argument bounds leakage by the commutator with the packet projector. Moving coordinate frames require a connection term; a canonical skew-Hermitian projector connection is constructed and shown to transport the complete packet family. The curvature of a projector bundle is expressed as $P(dP \wedge dP)P$, with the Bloch-sphere line bundle as a worked example. Measurements and completely positive maps are resolved into packet superoperators, and the Knill–Laflamme condition is interpreted as annihilation of the code obstruction packet. Exact Schur–Feshbach reduction proves that hidden sectors return as a spectral self-energy and, in time coordinates, as a memory kernel. The final sections formulate a finite projector geometry for quantum spacetime, prove positivity and composition of Euclidean transfer operators, test one- and two-dimensional spectral dimensions, and isolate the continuum, constraint, Lorentzian, Einstein-limit, and spin-two statements that remain conjectural. Nineteen code-generated figures, machine-readable results, and twenty-two automated tests accompany the paper. Finite theorems, classical inputs, numerical certificates, and conjectural physical claims are kept separate.

Scope and claim boundary. The paper proves the stated finite-dimensional results concerning quantum seed rank, exact packet reconstruction, cyclic symmetry projectors, tensor refinement, circuit path expansion, packet unitarity, leakage bounds, moving frames, projector connections, rank-one curvature, Kraus channels, block-diagonal measurement invariance, the finite Knill–Laflamme criterion, Schur–Feshbach reduction, and finite spectral diagnostics. It does not claim a solution of quantum gravity, a construction of an interacting continuum quantum field theory, anomaly-free gravitational constraints, Lorentzian reconstruction, the Einstein infrared limit, or the existence of a physical massless spin-two sector. Classical references are reserved for the consolidated end-of-series bibliography and are not reproduced here.

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1 The quantum seed realization

The common operator sequence of the series becomes

quantum seed $\longrightarrow$ gate or observable address $\longrightarrow$ symmetry projection $\longrightarrow$ packet transfer $\longrightarrow$ constraint $\longrightarrow$ refinement $\longrightarrow$ reconstruction

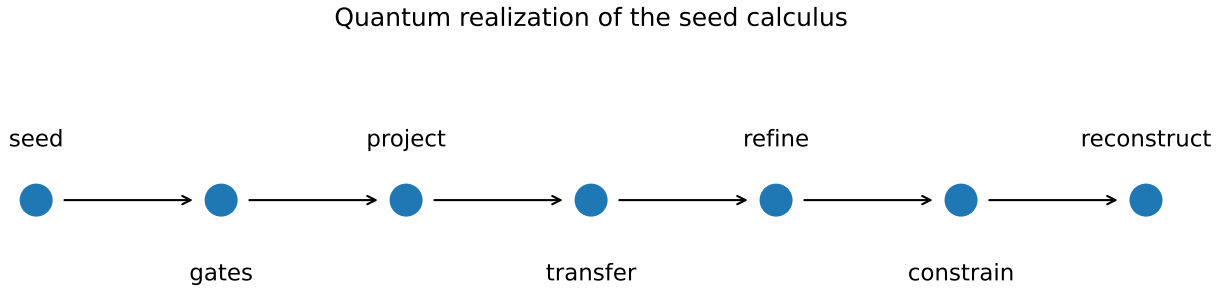


Figure 1: The quantum realization of the common seed-calculus operator chain. The same finite packet language is used for states, gates, Hamiltonians, channels, geometric projectors, and effective dynamics.

Definition 1.1 (Finite quantum seed system)**.** A finite quantum seed system is a tuple

$$\mathfrak{Q} = (\mathcal{H}, \psi_0, \mathcal{O}, \mathcal{R}, \{P_{\ell,\alpha}\}, H, \tau)\,,$$

where $\mathcal{H}$ is a finite-dimensional Hilbert space, $\psi_0 \in \mathcal{H}$ is a nonzero seed, $\mathcal{O}$ is the declared operator grammar, $\mathcal{R}$ is a collection of algebraic or physical constraints, $\{P_{\ell,\alpha}\}$ is an exact packet system at each refinement level, $H = H^*$ is a Hamiltonian when dynamics are present, and τ specifies the topology used in a later completion.

The generated state space is

$$\mathcal{H}(\psi_0; \mathcal{O}) = \operatorname{span}\{O_r \cdots O_1 \psi_0 : O_j \in \mathcal{O},\ r \geq 0\}\,.$$

The quantum seed is cyclic for the grammar when

$$\mathcal{H}(\psi_0; \mathcal{O}) = \mathcal{H}.$$

This is a statement about the pair $(\psi_0, \mathcal{O})$, not about the vector alone. A state may be cyclic for the full matrix algebra and noncyclic for a restricted physical gate set.

2 Quantum seed rank and cyclicity

2.1 Finite observable algebras

Let a finite-dimensional $*$-algebra act in its standard multiplicity form

$$\mathcal{A} \cong \bigoplus_{i=1}^{r} (M_{d_i}(\mathbb{C}) \otimes I_{m_i})$$

on

$$\mathcal{H} \cong \bigoplus_{i=1}^{r} \left(\mathbb{C}^{d_i} \otimes \mathbb{C}^{m_i} \right).$$

A vector $\psi_i \in \mathbb{C}^{d_i} \otimes \mathbb{C}^{m_i}$ is identified with a $d_i \times m_i$ coefficient matrix C_i.

Theorem 2.1 (Orbit dimension of one quantum seed)**.** *For the action of $M_d(\mathbb{C}) \otimes I_m$ on $\mathbb{C}^d \otimes \mathbb{C}^m$, let C be the coefficient matrix of a seed ψ. Then*

$$\boxed{\dim \operatorname{span}\{(A \otimes I_m)\psi : A \in M_d(\mathbb{C})\} = d \operatorname{rank} C.}$$

The seed is cyclic if and only if $\operatorname{rank} C = m$*, which is possible only when $m \leq d$.*

Proof. Under the coefficient-matrix identification, $(A \otimes I_m)\psi$ corresponds to AC. Every row of AC lies in the row space of C. Conversely, any $d \times m$ matrix whose rows lie in the row space of C equals AC for a suitable A: choose each row of A to produce the desired row combination. The orbit therefore consists of all matrices with d independently chosen rows in the $\operatorname{rank} C$-dimensional row space of C, and has dimension $d \operatorname{rank} C$. It equals the full dm-dimensional space exactly when $\operatorname{rank} C = m$. □

Theorem 2.2 (Quantum seed-rank formula)**.** *For*

$$\mathcal{A} = \bigoplus_i (M_{d_i}(\mathbb{C}) \otimes I_{m_i}) \quad \text{on} \quad \mathcal{H} = \bigoplus_i (\mathbb{C}^{d_i} \otimes \mathbb{C}^{m_i}),$$

the minimum number of seeds required to generate $\mathcal{H}$ is

$$\boxed{\operatorname{srank}_{\mathcal{A}}(\mathcal{H}) = \max_i \left\lceil \frac{m_i}{d_i} \right\rceil.}$$

Proof. Each seed contributes a coefficient matrix of rank at most d_i in the ith block. Thus k seeds can cover at most a kd_i-dimensional subspace of the multiplicity factor $\mathbb{C}^{m_i}$, so $k \geq \lceil m_i/d_i \rceil$ for every i.

For the converse, take

$$k = \max_i \left\lceil \frac{m_i}{d_i} \right\rceil.$$

In each block, partition a basis of $\mathbb{C}^{m_i}$ into at most k groups of size at most d_i. Choose the coefficient matrix of seed a so that its nonzero rows span the ath group. The combined row spaces of the k seeds then span $\mathbb{C}^{m_i}$ in every block. The preceding theorem shows that their $\mathcal{A}$-orbits generate the whole direct sum. □

Corollary 2.3 (Local orbit and Schmidt rank)**.** *Let $\psi \in \mathcal{H}_A \otimes \mathcal{H}_B$, with $\dim \mathcal{H}_A = d_A$, and let r be its Schmidt rank. Then*

$$\boxed{\dim \operatorname{span}\{(A \otimes I_B)\psi : A \in M_{d_A}(\mathbb{C})\} = d_A r.}$$

In particular, local operators on A generate the complete bipartite Hilbert space exactly when the reduced state on B has full support.

This gives a structural interpretation of entanglement within the seed calculus. Entanglement is not merely a scalar diagnostic; it controls how much of the remote multiplicity space is accessible from one seed under local addressing.

2.2 A computational-basis seed

Theorem 2.4 (One seed for the computational basis)**.** *For n qubits, let*

$$\psi_0 = |0\rangle^{\otimes n}$$

and let X_j denote the Pauli X operator on qubit j. Then

$$\{X_1^{a_1} \cdots X_n^{a_n} \psi_0 : (a_1, \ldots, a_n) \in \{0,1\}^n\}$$

is the complete computational basis. Hence the seed is cyclic under the algebra generated by the X_j.

Proof. For a binary address $a = (a_1, \ldots, a_n)$,

$$X_1^{a_1} \cdots X_n^{a_n} |0 \cdots 0\rangle = |a_1 \cdots a_n\rangle .$$

Distinct binary addresses produce distinct orthonormal basis vectors. There are 2^n of them, equal to the dimension of the Hilbert space. □

Numerical Certificate 2.5 (Three-qubit seed orbit). The implementation generated all eight computational basis states from $|000\rangle$ using the three local X address operators. The measured orbit rank was 8/8.

3 Cyclic symmetry packets

3.1 Character projectors

Let S be a unitary symmetry of finite order N:

$$S^N = I.$$

Set $\omega = e^{2\pi i/N}$.

Theorem 3.1 (Cyclic quantum projectors)**.** *The operators*

$$\boxed{P_j = \frac{1}{N} \sum_{r=0}^{N-1} \omega^{-jr} S^r, \qquad 0 \le j < N,}$$

are pairwise orthogonal projectors satisfying

$$P_j P_k = \delta_{jk} P_j, \qquad \sum_{j=0}^{N-1} P_j = I, \qquad S P_j = \omega^j P_j.$$

Proof. Using $S^{r+s} = S^{r+s \bmod N}$ and the finite character orthogonality relation,

$$\frac{1}{N} \sum_{r=0}^{N-1} \omega^{(k-j)r} = \delta_{jk},$$

one obtains $P_j P_k = \delta_{jk} P_j$. Summing over j annihilates every nonzero power of S and retains the identity term, proving completeness. Finally, multiplying by S shifts the summation index and contributes the factor ω^j. □

For parity,

$$S = Z^{\otimes n}, \qquad P_{\text{even}} = \frac{I+S}{2}, \qquad P_{\text{odd}} = \frac{I-S}{2}.$$

3.2 Tensor refinement

Suppose factor a has projectors $P_0^{(a)}, \dots, P_{N-1}^{(a)}$. The fine tensor packets are

$$P_{\mathbf{j}}^{\text{fine}} = P_{j_1}^{(1)} \otimes \cdots \otimes P_{j_m}^{(m)}.$$

Theorem 3.2 (Tensor/polyphase refinement)**.** *For each residue $r \in \mathbb{Z}/N\mathbb{Z}$, define*

$$\boxed{P_r^{\text{coarse}} = \sum_{j_1+\cdots+j_m \equiv r \pmod N} P_{j_1}^{(1)} \otimes \cdots \otimes P_{j_m}^{(m)}.}$$

Then the P_r^{coarse} are complete orthogonal projectors, and every coarse packet is the exact sum of its fine descendants.

Proof. The fine tensor projectors are mutually orthogonal and sum to the identity. The stated coarse families partition their address set by total residue. Sums over disjoint sets of orthogonal projectors remain orthogonal projectors, and the union of the residue classes is the full fine address set. □

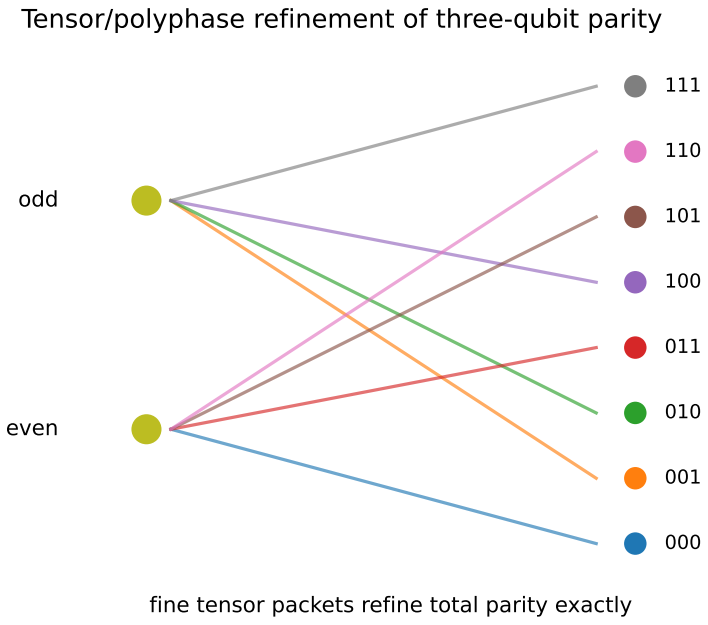


Figure 2: The eight computational packets of three qubits refine the two total-parity packets exactly.

Numerical Certificate 3.3 (Exact parity refinement). For three qubits, the local computational projectors were grouped by Hamming-weight parity. The resulting coarse projectors agreed with $(I \pm Z^{\otimes 3})/2$ with residual 0.

4 Quantum circuits as packet paths

Let

$$U = U_m U_{m-1} \cdots U_1$$

be a circuit and let $\{P_\alpha\}$ be a complete packet system.

Theorem 4.1 (Exact circuit path expansion)**.** *For source packet α_0 and target packet α_m,*

$$\boxed{P_{\alpha_m} U P_{\alpha_0} = \sum_{\alpha_1,\dots,\alpha_{m-1}} P_{\alpha_m} U_m P_{\alpha_{m-1}} \cdots P_{\alpha_1} U_1 P_{\alpha_0}.}$$

Summing over the source and target packets reconstructs the complete circuit operator.

Proof. Insert the identity resolution $I = \sum_\alpha P_\alpha$ between every adjacent pair of gates. Distributing the finite sums gives the path expansion. Summing the left and right boundary projectors reconstructs U. □

Theorem 4.2 (Packetwise unitarity conservation)**.** *If U is unitary and $U_{\beta\alpha} = P_\beta U P_\alpha$, then*

$$\boxed{\sum_\beta U^*_{\beta\alpha} U_{\beta\gamma} = \delta_{\alpha\gamma} P_\alpha.}$$

Similarly,

$$\sum_\alpha U_{\beta\alpha} U^*_{\delta\alpha} = \delta_{\beta\delta} P_\beta.$$

Proof. Using completeness and unitarity,

$$\sum_\beta U^*_{\beta\alpha} U_{\beta\gamma} = P_\alpha U^* \left(\sum_\beta P_\beta \right) U P_\gamma = P_\alpha P_\gamma.$$

The second identity is analogous. □

Example 4.3 (Bell seed circuit)**.** Starting from $|00\rangle$,

$$(H \otimes I)\,|00\rangle = \frac{|00\rangle + |10\rangle}{\sqrt{2}},$$

and then

$$\mathrm{CNOT}(H \otimes I)\,|00\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

The parity packets show that the final state lies entirely in the even sector, while the intermediate circuit path passes through both even and odd addresses.

Numerical Certificate 4.4 (Bell and path reconstruction)*.* The Bell-state residual and the complete parity-path reconstruction residual were both 0 in the numerical implementation.

5 Symmetry-resolved Hamiltonians and leakage

Let $H = H^*$ and let P be a packet projector. If $[H, P] = 0$, then $P\mathcal{H}$ is invariant under e^{-itH}.

Theorem 5.1 (Exact symmetry packets)**.** *If $[H, P_\alpha] = 0$ for every packet, then*

$$H = \sum_\alpha P_\alpha H P_\alpha$$

and

$$e^{-itH} = \sum_\alpha P_\alpha e^{-itH} P_\alpha.$$

No amplitude transfers between distinct packets.

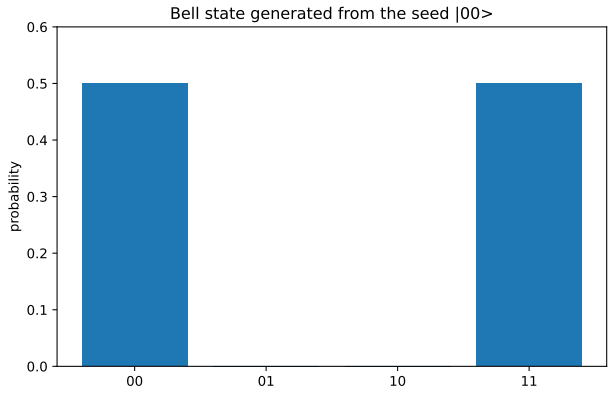


Figure 3: The Bell state generated from the seed $|00\rangle$.

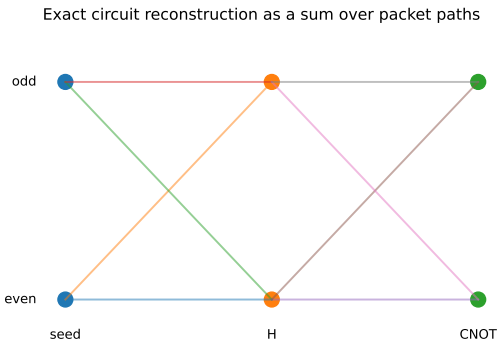


Figure 4: Circuit reconstruction as a sum over packet paths.

Proof. The commutation relation gives $P_\beta H P_\alpha = 0$ for $\beta \neq \alpha$. Every polynomial in H has the same block form, and the finite-dimensional exponential is the norm limit of its polynomial series. □

For two qubits, consider

$$H = JX \otimes X + h(Z \otimes I + I \otimes Z).$$

In the ordered even basis $\{|00\rangle, |11\rangle\}$ and odd basis $\{|01\rangle, |10\rangle\}$,

$$H_{\text{even}} = \begin{pmatrix} 2h & J \\ J & -2h \end{pmatrix}, \qquad H_{\text{odd}} = \begin{pmatrix} 0 & J \\ J & 0 \end{pmatrix}.$$

Therefore

$$\operatorname{Spec} H_{\text{even}} = \{\pm\sqrt{4h^2 + J^2}\}, \qquad \operatorname{Spec} H_{\text{odd}} = \{\pm J\}.$$

5.1 Leakage under broken symmetry

Set $Q = I - P$.

Theorem 5.2 (Commutator leakage bound)**.** *For every self-adjoint H, orthogonal projector P, and real time t,*

$$\boxed{\left\| Q e^{-itH} P \right\| \leq |t| \, \|[H, P]\|.}$$

In particular, exact commutation gives zero leakage for all time.

Proof. Because $QP = 0$,

$$Q e^{-itH} P = Q[e^{-itH}, P]P.$$

The Duhamel identity gives

$$[e^{-itH}, P] = -i \int_0^t e^{-i(t-s)H} [H, P] e^{-isH} \, ds.$$

Both unitary factors have norm one, so taking norms yields the bound. □

Numerical Certificate 5.3 (Symmetry and negative control). For the parity-preserving Hamiltonian, the off-packet block norm and maximum measured leakage were both 0. After adding a parity-breaking $X \otimes I$ perturbation, the maximum leakage was approximately 0.79196, and the maximum violation of the proved bound was 0.

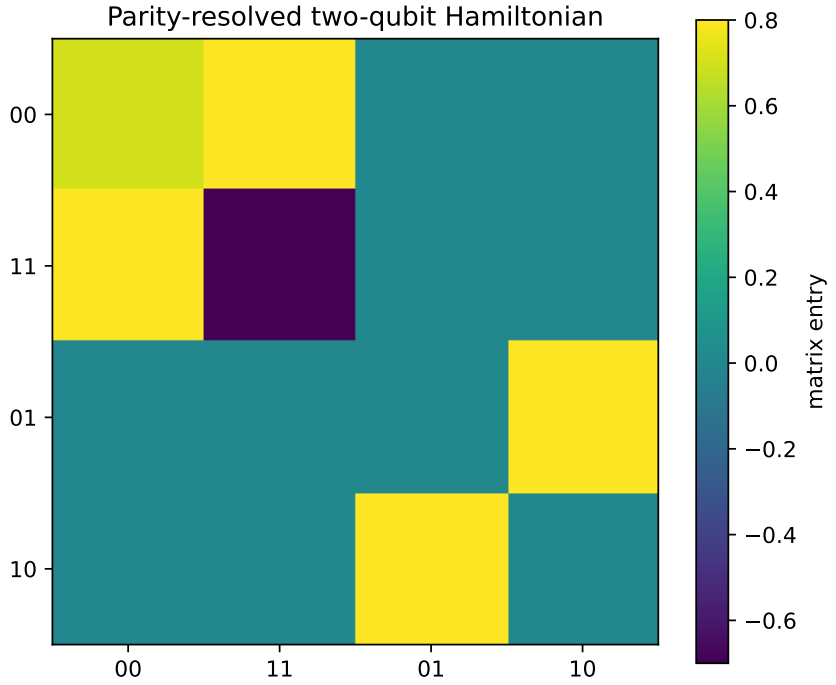


Figure 5: The parity-resolved Hamiltonian is exactly block diagonal after the even and odd basis vectors are grouped.

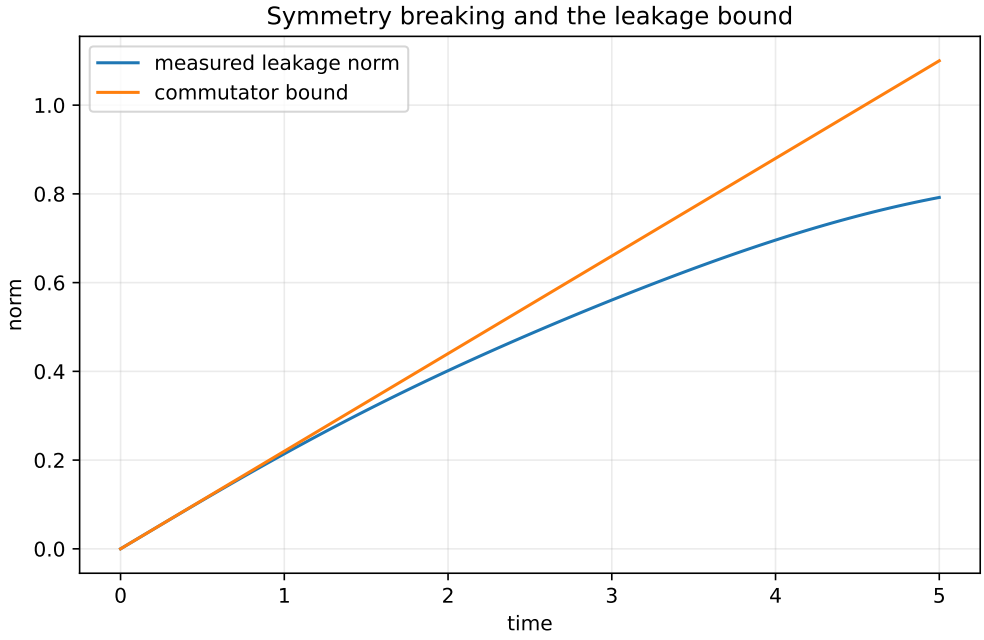


Figure 6: A parity-breaking perturbation creates nonzero transfer between the packets, while the measured leakage remains below the commutator bound.

6 Moving packets, connections, and curvature

6.1 Moving quantum coordinates

Theorem 6.1 (Moving-frame Hamiltonian)**.** *Let $U(t)$ be differentiable and unitary, and suppose*

$$i\dot{\psi}(t) = H(t)\psi(t).$$

Writing

$$\psi(t) = U(t)y(t)$$

gives

$$\boxed{i\dot{y}(t) = \Big(U(t)^* H(t)U(t) - iU(t)^*\dot{U}(t)\Big)\, y(t).}$$

The connection term $-iU^*\dot{U}$ is Hermitian.

Proof. Differentiate $\psi = Uy$:

$$i\dot{U}y + iU\dot{y} = HUy.$$

Multiplying by U^* gives the formula. Differentiating $U^*U = I$ shows $U^*\dot{U}$ is skew-Hermitian, so $-iU^*\dot{U}$ is Hermitian. □

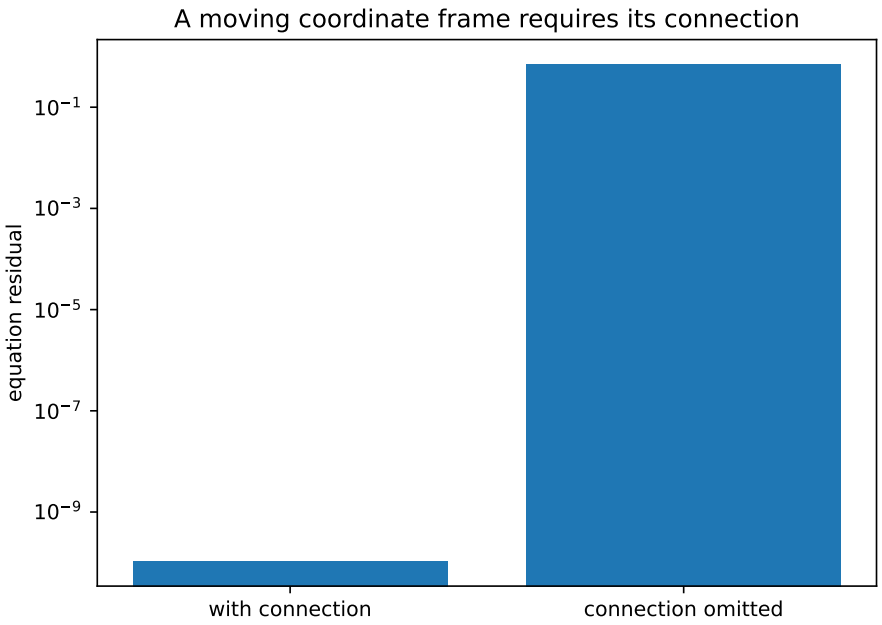


Figure 7: The moving-frame equation closes to numerical precision only when the connection term is retained.

Numerical Certificate 6.2 (Moving-frame correction). The complete moving-frame equation had residual 1.0645×10^{-10}, while omitting the connection produced residual 0.7000.

6.2 A canonical projector connection

Let $P_1(t), \ldots, P_m(t)$ be differentiable orthogonal projectors satisfying

$$P_a P_b = \delta_{ab} P_a, \qquad \sum_a P_a = I.$$

Theorem 6.3 (Canonical packet connection)**.** *Define*

$$\boxed{C(t) = \frac{1}{2}\sum_a [\dot{P}_a(t), P_a(t)] = \sum_a \dot{P}_a(t) P_a(t).}$$

Then

$$C(t)^* = -C(t)$$

and

$$\boxed{[C(t), P_b(t)] = \dot{P}_b(t)}$$

for every packet b.

Proof. Differentiating $P_a^2 = P_a$ gives

$$\dot{P}_a P_a + P_a \dot{P}_a = \dot{P}_a.$$

Summing over a and using $\sum_a \dot{P}_a = 0$ yields

$$\sum_a P_a \dot{P}_a = -\sum_a \dot{P}_a P_a,$$

which proves both displayed formulas for C and its skew-Hermiticity.

For fixed b,

$$C P_b = \dot{P}_b P_b.$$

When $a \neq b$, differentiating $P_b P_a = 0$ gives

$$P_b \dot{P}_a = -\dot{P}_b P_a.$$

Hence

$$P_b C = \sum_a P_b \dot{P}_a P_a = -\dot{P}_b (I - P_b),$$

and subtraction gives $[C, P_b] = \dot{P}_b$. □

The connection is the infinitesimal transport operator of the moving packet frame. It is determined by the packet family itself, without choosing a basis inside any packet.

6.3 Projector curvature

Classical Input 6.4 (Projected connection). For a smooth projector-valued map P on a parameter manifold, the connection on the range bundle is $\nabla = P\, d$. Its curvature is

$$\boxed{F = P(dP \wedge dP)P.}$$

Proof. For a section s with $Ps = s$,

$$\nabla^2 s = P\, d(P\, ds) = P(dP) \wedge ds.$$

Since $ds = d(Ps) = (dP)s + P ds$ and $P(dP)P = 0$, the surviving term is

$$P(dP \wedge dP)Ps.$$

□

For a rank-one projector, the scalar curvature density is

$$\mathcal{F}_{\mu\nu} = -i \operatorname{Tr}\left(P[\partial_\mu P, \partial_\nu P]\right).$$

On the Bloch sphere,

$$P(\theta,\phi) = \frac{1}{2}\left(I + \mathbf{n}(\theta,\phi)\cdot\boldsymbol{\sigma}\right),$$

where

$$\mathbf{n} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta).$$

The Pauli multiplication law gives

$$\boxed{\mathcal{F}_{\theta\phi} = \frac{1}{2}\sin\theta.}$$

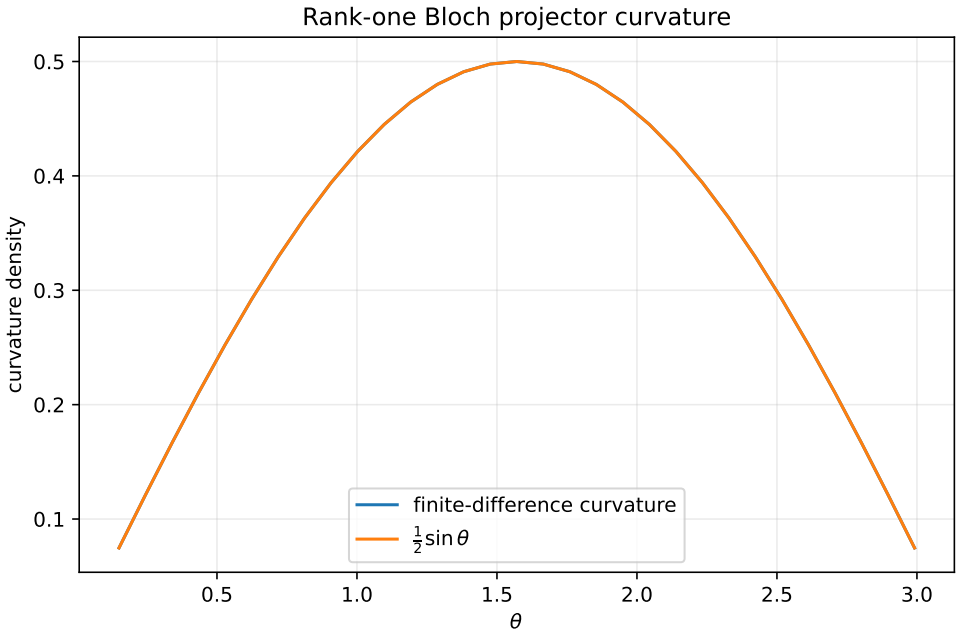


Figure 8: Finite-difference evaluation of the Bloch projector curvature against the exact density $\frac{1}{2}\sin\theta$.

Numerical Certificate 6.5 (Geometric packet identities). The canonical connection reconstructed every packet derivative with residual 0. The maximum Bloch-curvature residual was 2.0803×10^{-11}.

7 Measurements and quantum channels

7.1 Packet measurements

Let ρ be a density matrix and let $\{E_k\}$ be a POVM:

$$E_k \geq 0, \qquad \sum_k E_k = I.$$

The outcome probabilities are

$$p_k = \operatorname{Tr}(E_k\rho).$$

Theorem 7.1 (Block-diagonal measurement invariance)**.** *Let $\{P_\alpha\}$ be a complete packet family, and suppose every measurement effect is packet diagonal:*

$$E_k = \sum_\alpha P_\alpha E_k P_\alpha.$$

Define the packet-dephased state

$$\mathcal{D}_P(\rho) = \sum_\alpha P_\alpha \rho P_\alpha.$$

Then

$$\boxed{\mathrm{Tr}(E_k\rho) = \mathrm{Tr}(E_k \mathcal{D}_P(\rho))}$$

for every outcome k.

Proof. Insert the packet decomposition of E_k and use cyclicity of the trace:

$$\mathrm{Tr}(E_k\rho) = \sum_\alpha \mathrm{Tr}(P_\alpha E_k P_\alpha \rho) = \sum_\alpha \mathrm{Tr}(E_k P_\alpha \rho P_\alpha).$$

□

This theorem says exactly what coherence a declared measurement can and cannot see. Coherence between different packets is invisible to every packet-diagonal observable, while coherence inside a packet remains available.

7.2 Kraus transfer

Classical Input 7.2 (Finite Kraus channel). Let $K_1, \ldots, K_r$ satisfy

$$\sum_a K_a^* K_a = I.$$

Then

$$\boxed{\Phi(\rho) = \sum_a K_a \rho K_a^*}$$

is completely positive and trace preserving.

Proof. Trace preservation follows from

$$\mathrm{Tr}\,\Phi(\rho) = \mathrm{Tr}\left(\rho \sum_a K_a^* K_a\right) = \mathrm{Tr}\,\rho.$$

For any auxiliary dimension and positive matrix X, each term

$$(I \otimes K_a) X (I \otimes K_a)^*$$

is positive, so their sum is positive. This is complete positivity. □

Packet superoperator blocks are defined by

$$\Phi_{\alpha\beta;\gamma\delta}(X) = P_\alpha \Phi(P_\gamma X P_\delta) P_\beta.$$

Summing all four packet indices reconstructs $\Phi(X)$ exactly.

For amplitude damping with parameter γ,

$$K_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix}, \qquad K_1 = \begin{pmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{pmatrix}.$$

The excited state becomes

$$\Phi(|1\rangle\langle 1|) = \gamma\,|0\rangle\langle 0| + (1-\gamma)\,|1\rangle\langle 1|\,.$$

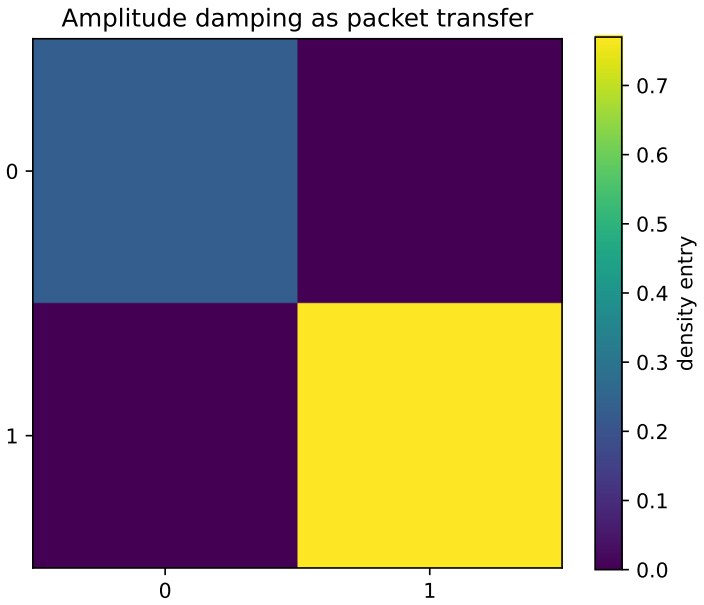


Figure 9: Amplitude damping transfers population from the excited packet to the ground packet while preserving trace and positivity.

Numerical Certificate 7.3 (Channel reconstruction). For $\gamma = 0.23$, the Kraus completeness and output-trace residuals were 0. The computed output from $|1\rangle\langle 1|$ was $\mathrm{diag}(0.23, 0.77)$. Packet dephasing changed no computational-basis measurement probability in the Bell-state test.

8 Quantum error correction as obstruction-packet annihilation

Let P project onto a code space $\mathcal{C} \subset \mathcal{H}$, and let $E_1, \ldots, E_r$ be an error family. The operators

$$PE_a^* E_b P$$

are the code-level obstruction packets: if they retain logical-state information beyond one scalar multiple of P, the environment can distinguish logical states and exact correction is impossible.

Classical Input 8.1 (Finite Knill–Laflamme criterion). The error family is exactly correctable on $\mathcal{C}$ if and only if there is a positive semidefinite matrix (c_{ab}) such that

$$\boxed{PE_a^* E_b P = c_{ab} P}$$

for every a, b.

Proof. Assume first that a recovery corrects every error combination. Then inner products between corrupted code states cannot depend on the logical vector except through its original inner product. Polarization therefore gives

$$\langle\psi| E_a^* E_b |\phi\rangle = c_{ab}\langle\psi, \phi\rangle$$

for all $\psi, \phi \in \mathcal{C}$, which is the displayed operator identity.

Conversely, diagonalize the positive matrix c and replace the E_a by the corresponding linear combinations F_r. Then

$$PF_r^* F_s P = \lambda_r \delta_{rs} P.$$

For $\lambda_r > 0$, $F_r/\sqrt{\lambda_r}$ is an isometry from the code into an error subspace, and distinct error subspaces are orthogonal. A recovery first projects onto these syndrome packets and then applies the inverse isometry on each packet. This reconstructs every code state. □

For the three-qubit bit-flip code,

$$|0_L\rangle = |000\rangle\,, \qquad |1_L\rangle = |111\rangle\,,$$

and the error family is

$$I, \quad X \otimes I \otimes I, \quad I \otimes X \otimes I, \quad I \otimes I \otimes X.$$

The four error images of the code are pairwise orthogonal two-dimensional syndrome packets whose projectors sum to the identity on the eight-dimensional Hilbert space.

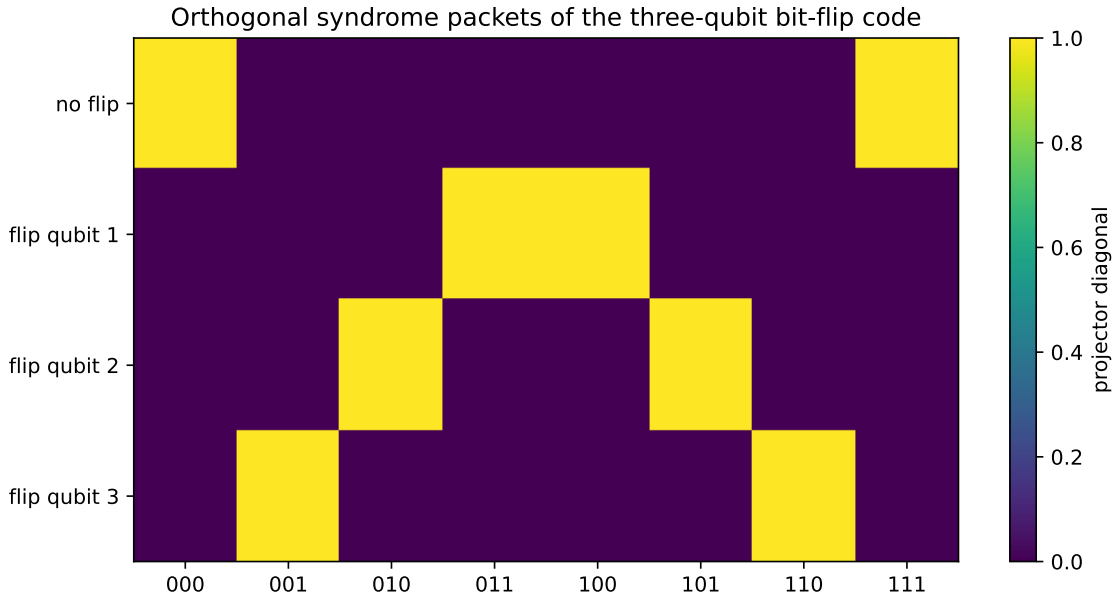


Figure 10: The four orthogonal syndrome packets of the three-qubit bit-flip code. Each packet identifies one correctable error without resolving the encoded logical state.

Numerical Certificate 8.2 (Exact bit-flip correction). The Knill–Laflamme residual and syndrome-completeness residual were both 0. A mixed single-bit-flip channel applied to a nontrivial logical superposition was reversed with density-matrix residual 1.0385×10^{-16}.

9 Exact effective dynamics

9.1 Schur–Feshbach reduction

Let P be a retained packet projector and $Q = I - P$. In packet coordinates,

$$H = \begin{pmatrix} H_{PP} & H_{PQ} \\ H_{QP} & H_{QQ} \end{pmatrix}.$$

Theorem 9.1 (Exact Feshbach resolvent). *If $z - H_{QQ}$ is invertible, define*

$$\boxed{H_{\mathrm{eff}}(z) = H_{PP} + H_{PQ}(z - H_{QQ})^{-1} H_{QP}.}$$

Then

$$\boxed{P(z-H)^{-1}P=(z-H_{\mathrm{eff}}(z))^{-1}}$$

when both sides are written in an orthonormal basis of $P\mathcal{H}$.

Proof. The block matrix $z-H$ has lower-right block $z-H_{QQ}$. Gaussian block elimination produces the Schur complement

$$z-H_{PP}-H_{PQ}(z-H_{QQ})^{-1}H_{QP}.$$

The upper-left block of the inverse is the inverse of this Schur complement, which is exactly the displayed operator. □

The self-energy

$$\Sigma_P(z)=H_{PQ}(z-H_{QQ})^{-1}H_{QP}$$

contains the complete influence of the eliminated packet. Bare truncation drops $\Sigma_P(z)$ and is therefore exact only when the cross-packet coupling vanishes or the self-energy cancels for another stated reason.

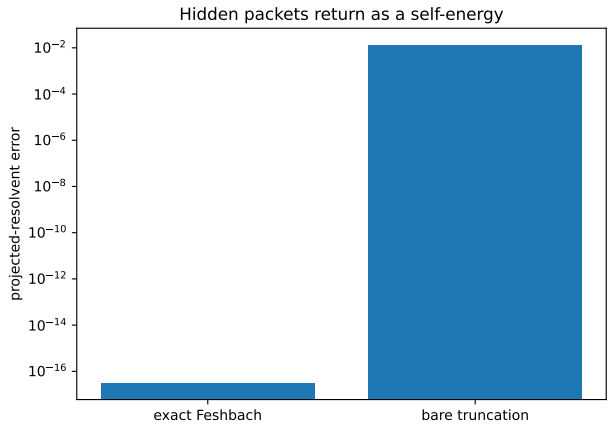


Figure 11: Exact Feshbach reduction versus bare truncation.

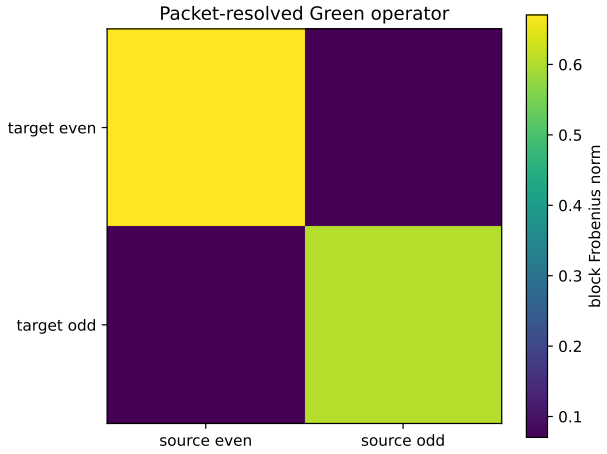


Figure 12: Norms of packet Green-function blocks.

Numerical Certificate 9.2 (Effective resolvent). For a random six-dimensional Hermitian test, the exact Feshbach residual was 3.2291×10^{-17}. The corresponding bare-truncation residual was 1.3091×10^{-2}.

9.2 The hidden packet as memory

Write $\psi=x+y$, with $x=P\psi$ and $y=Q\psi$. Then

$$i\dot{x}=H_{PP}x+H_{PQ}y,$$
$$i\dot{y}=H_{QP}x+H_{QQ}y.$$

Solving the second equation gives

$$y(t)=e^{-itH_{QQ}}y(0)-i\int_0^t e^{-i(t-s)H_{QQ}}H_{QP}x(s)\,ds.$$

Substitution yields the exact Volterra equation

$$\boxed{\begin{aligned}i\dot{x}(t)&=H_{PP}x(t)+H_{PQ}e^{-itH_{QQ}}y(0)\\&\quad-i\int_0^t K(t-s)x(s)\,ds,\end{aligned}}$$

with memory kernel

$$\boxed{K(t) = H_{PQ} e^{-itH_{QQ}} H_{QP}.}$$

Thus hidden packets do not disappear. They become a frequency-dependent self-energy in resolvent coordinates and a causal memory kernel in time coordinates.

10 Finite quantum geometry

10.1 A projector-resolved geometry

At resolution ℓ, define

$$\boxed{\mathfrak{Q}_\ell = \left(\mathcal{A}_\ell, \mathcal{H}_\ell, D_\ell, \{P_{\ell,\alpha}\}, E_\ell, C_\ell, \Pi_\ell^{\mathrm{phys}}, T_\ell\right).}$$

The entries are an observable algebra, Hilbert space, geometric or Dirac-type operator, packet projectors, fine embedding, coarse reconstruction, physical constraint projector, and transfer operator.

A coherent refinement should satisfy

$$C_\ell E_\ell = I$$

and

$$\boxed{P_{\ell,\alpha} = C_\ell \left(\sum_{\beta \succ \alpha} P_{\ell+1,\beta} \right) E_\ell.}$$

For an operator D, define the refinement defect

$$\Delta_\ell(D) = C_\ell D_{\ell+1} E_\ell - D_\ell.$$

Exact resolution independence is $\Delta_\ell(D) = 0$. When exact equality is too strong, the correct continuum problem is to prove convergence of the resolvents and uniform control of the effective self-energies.

10.2 Euclidean transfer

Theorem 10.1 (Finite Euclidean transfer)**.** *For every finite Hermitian H bounded below and every $\tau \geq 0$,*

$$T(\tau) = e^{-\tau(H - E_0 I)}, \qquad E_0 = \min \operatorname{Spec} H,$$

is positive definite and satisfies

$$\boxed{T(\tau + \sigma) = T(\tau) T(\sigma).}$$

Its spectrum is contained in $(0, 1]$.

Proof. Diagonalize $H = \sum_j E_j P_j$. Then

$$T(\tau) = \sum_j e^{-\tau(E_j - E_0)} P_j.$$

Every eigenvalue is positive and at most one. Multiplication adds the exponents, proving the semigroup law. □

The paper tests a controlled volume-state model

$$H_{nn} = 2\kappa + V(n), \qquad H_{n,n+1} = H_{n+1,n} = -\kappa,$$

with a confining potential. This is a finite transfer-matrix test only. It is not derived from a gravitational path integral and is not evidence for a unique microscopic spacetime dynamics.

Numerical Certificate 10.2 (Finite transfer model). The minimum transfer eigenvalue was 0.0767054, the semigroup residual was 0, and reconstruction from four volume-residue packets had residual 0.

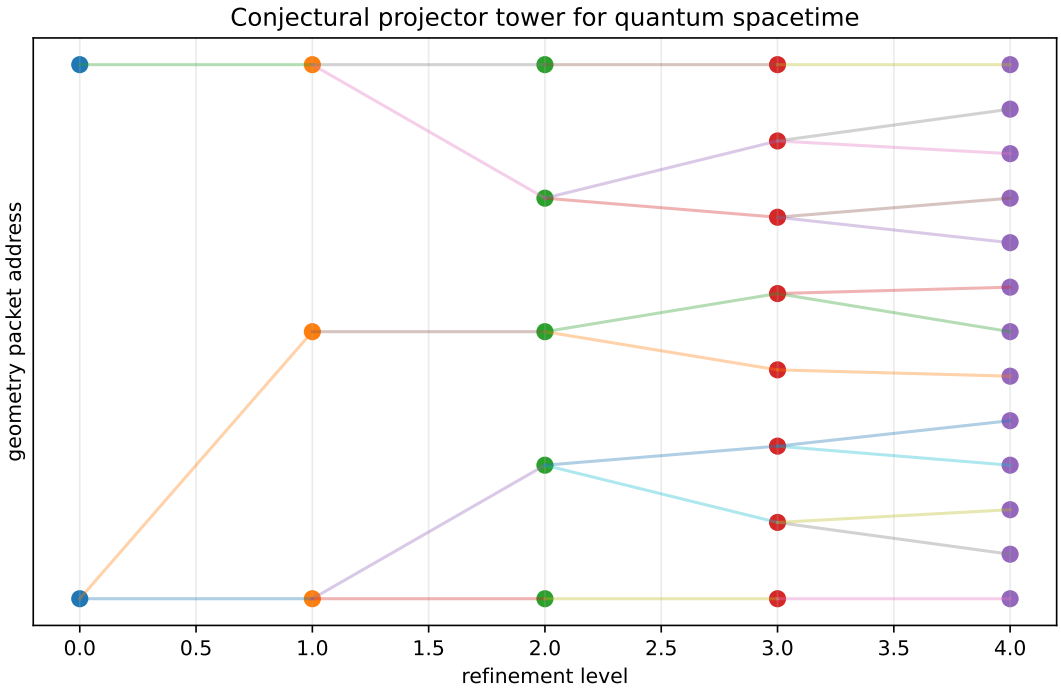


Figure 13: A conjectural nested packet tower for quantum geometry. The diagram is an organizational model; it is not a numerical reconstruction of physical spacetime.

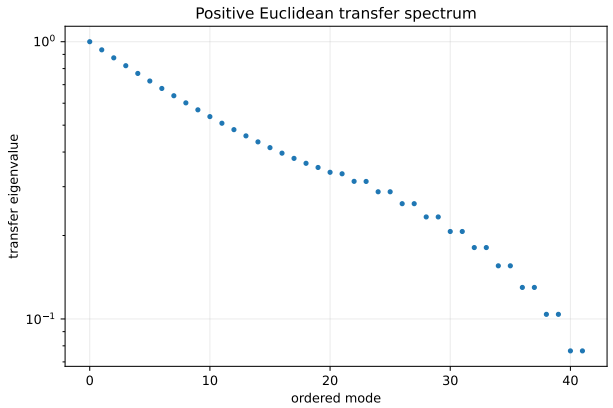


Figure 14: Positive spectrum of the Euclidean transfer operator.

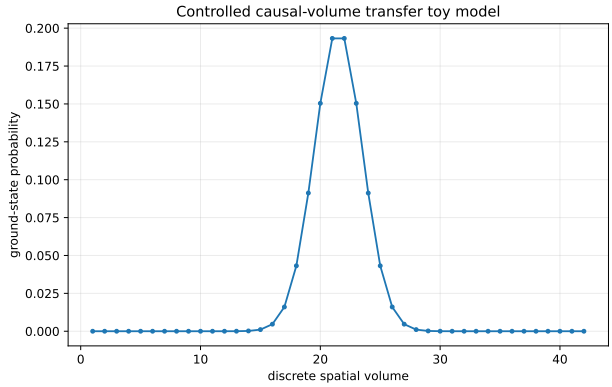


Figure 15: Ground-state volume packet in the controlled toy model.

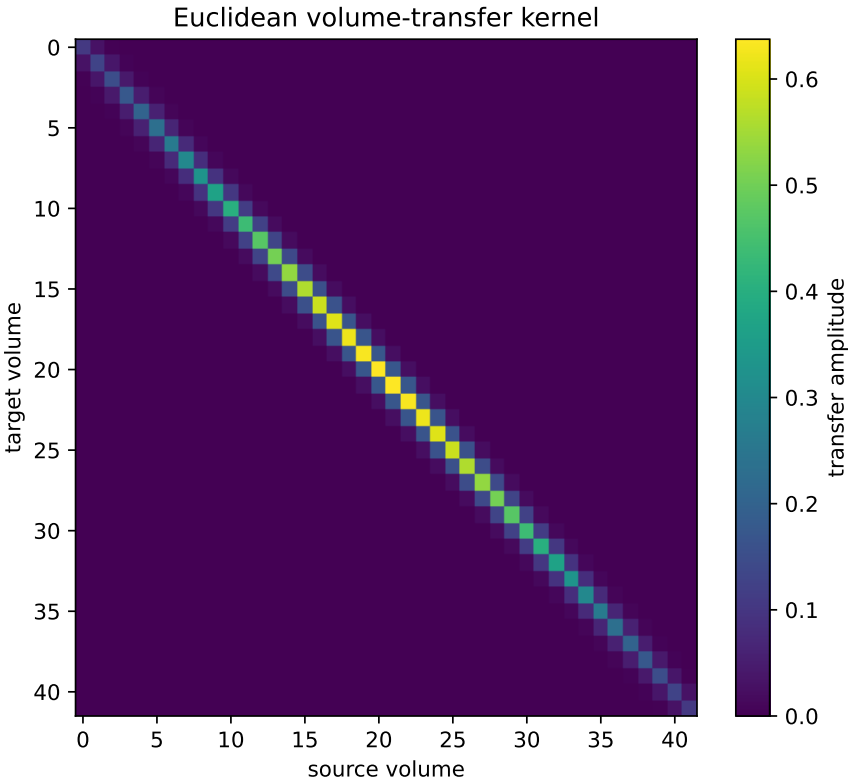


Figure 16: The finite Euclidean volume-transfer kernel. Positivity and semigroup composition are exact finite statements; its gravitational interpretation remains provisional.

11 Spectral dimension and continuum diagnostics

Let $L \geq 0$ be a finite Laplacian with eigenvalues λ_j. Define the normalized heat trace

$$K(\tau) = \frac{1}{N} \sum_j e^{-\tau \lambda_j}$$

and the scale-dependent spectral dimension

$$\boxed{d_s(\tau) = -2 \frac{d \log K(\tau)}{d \log \tau}.}$$

For a finite system, $d_s(\tau)$ is a diagnostic, not a topological invariant. At very small diffusion time it sees the ultraviolet lattice, and at very large time it collapses toward the zero mode. A geometric dimension can only be inferred from a stable intermediate window and a controlled refinement limit.

11.1 The circle dispersion bound

For a periodic grid with spacing h,

$$\lambda_{k,h} = \frac{4}{h^2} \sin^2 \left(\frac{kh}{2} \right).$$

Theorem 11.1 (Low-mode second-order bound)**.** *If* $0 \leq kh/2 \leq 1$*, then*

$$\boxed{0 \leq k^2 - \lambda_{k,h} \leq \frac{k^4 h^2}{12}.}$$

Proof. Set $x = kh/2$. Since $\sin x \leq x$, the error is nonnegative. On $[0, 1]$,

$$\sin x \geq x - \frac{x^3}{6}.$$

Therefore

$$\sin^2 x \geq \left(x - \frac{x^3}{6} \right)^2 \geq x^2 - \frac{x^4}{3}.$$

Hence

$$x^2 - \sin^2 x \leq \frac{x^4}{3}.$$

Multiplication by $4/h^2$ gives the result. □

Numerical Certificate 11.2 (Continuum dispersion). The observed convergence order was 1.99936. The maximum error-minus-bound value was negative, so no tested mode violated the theorem.

11.2 One- and two-dimensional plateaus

Numerical Certificate 11.3 (Spectral dimension). The median intermediate plateaus were

$$d_s^{(1)} = 1.00062, \qquad d_s^{(2)} = 2.03781.$$

These tests recover the expected dimensions of the chosen finite geometries; they do not select a physical quantum-spacetime theory.

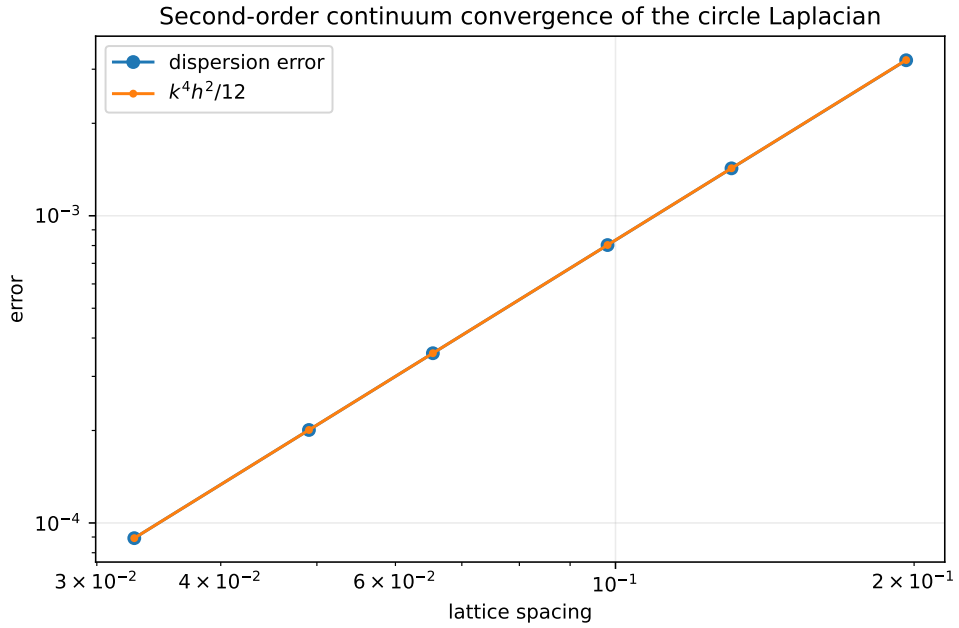


Figure 17: The first nonzero circle mode converges at second order and remains below the proved dispersion bound.

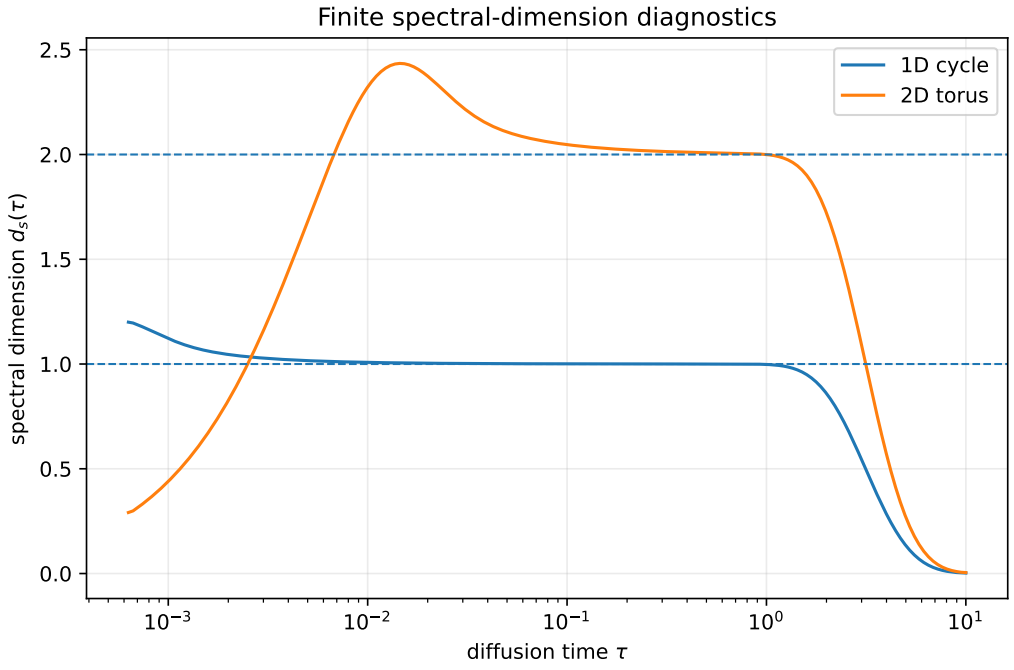


Figure 18: Intermediate spectral-dimension windows for a finite cycle and a finite two-dimensional torus. The ultraviolet and infrared departures are finite-size effects, not additional dimensions.

12 Constraint packets and the quantum-spacetime boundary

Let $\widehat{C}_A$ be quantum constraints and let Π^{phys} project onto their common kernel. If the classical constraint algebra has structure coefficients $f_{AB}^{\ C}$, define the anomaly operator

$$\boxed{\mathfrak{A}_{AB} = [\widehat{C}_A, \widehat{C}_B] - i \sum_C f_{AB}^{\ C} \widehat{C}_C.}$$

Its physical packet is

$$\Pi^{\mathrm{phys}} \mathfrak{A}_{AB} \Pi^{\mathrm{phys}}.$$

A nonzero physical anomaly packet is an explicit obstruction to exact constraint closure.

Conjecture 12.1 (Anomaly-free refinement)**.** There exists a coherent quantum-geometric packet tower for which

$$\Pi_\ell^{\mathrm{phys}} \mathfrak{A}_{AB}^{(\ell)} \Pi_\ell^{\mathrm{phys}} = 0$$

at every level and the physical projectors reconstruct compatibly under refinement.

Conjecture 12.2 (Uniform Feshbach control)**.** For the same tower, the self-energies produced by exact elimination of fine packets admit bounds uniform in the cutoff and volume on a common spectral domain. These bounds yield a strong-resolvent continuum limit.

Conjecture 12.3 (Lorentzian reconstruction)**.** The Euclidean packet theory satisfies a positivity or reconstruction principle strong enough to produce a Lorentzian Hilbert space with causal propagation and positive physical norm.

Conjecture 12.4 (Einstein infrared limit)**.** There exist semiclassical packet states $\Psi_{g,\ell}$ such that the effective action converges to

$$\frac{1}{16\pi G} \int \sqrt{-g}\,(R - 2\Lambda)\, d^4x$$

with controlled higher-order corrections.

Conjecture 12.5 (Massless spin-two sector)**.** The continuum two-point function possesses a positive transverse-traceless massless spin-two pole, while all unphysical packet sectors are removed by the constraints.

These conjectures are logically independent of the exact finite identities. Exact packet reconstruction is a kinematic and computational advantage; it does not determine the physical action, prove continuum existence, or guarantee the general-relativistic limit.

13 Numerical certificates

All experiments in this paper were run on CPU. No GPU result is claimed.

The complete automated suite passed

$$\boxed{22/22 \text{ tests.}}$$

The results are stored in machine-readable JSON and CSV form. Every figure in the paper is generated by the accompanying Python package.

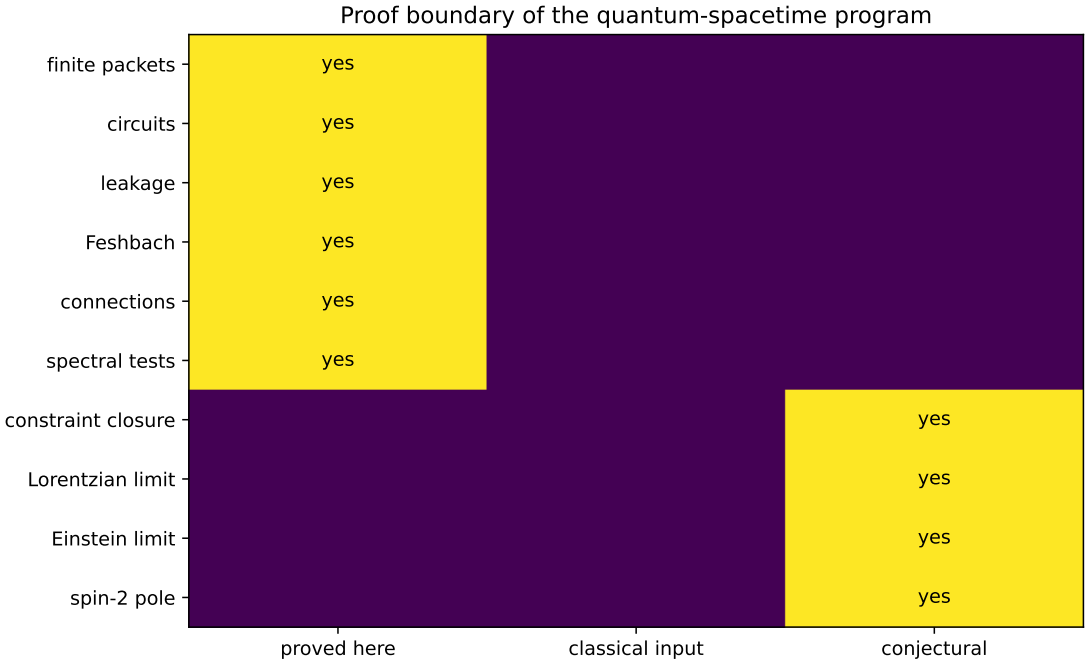


Figure 19: The proof boundary. Finite packet, circuit, leakage, Feshbach, connection, and spectral identities are established here. Constraint closure and the physical continuum claims remain conjectural.

Table 1: Representative finite numerical certificates.

Certificate	Value
Bell-state residual	0
Three-qubit seed-orbit rank	8/8
Parity off-block norm	0
Circuit path reconstruction residual	0
Exact-symmetry maximum leakage	0
Maximum leakage-bound violation	0
Feshbach projected-resolvent residual	3.2291×10^{-17}
Bare-truncation projected-resolvent residual	1.3091×10^{-2}
Moving-frame residual with connection	1.0645×10^{-10}
Moving-frame residual without connection	7.0000×10^{-1}
Canonical packet-connection residual	0
Bloch-curvature residual	2.0803×10^{-11}
Kraus completeness residual	0
Output trace residual	0
Knill–Laflamme residual	0
Bit-flip recovery residual	1.0385×10^{-16}
Tensor parity-refinement residual	0
Circle low-mode convergence order	1.99936
One-dimensional spectral plateau	1.00062
Two-dimensional spectral plateau	2.03781
Minimum Euclidean-transfer eigenvalue	0.0767054
Transfer semigroup residual	0
Volume-packet reconstruction residual	0

14 Software realization

The companion implementation is organized by mathematical responsibility:

Module	Responsibility
`basic.py`	basis states, Pauli gates, tensor embeddings, packet blocks, cyclic and multi-generator reachability
`symmetry.py`	finite cyclic projectors, parity packets, tensor refinement, sector populations
`circuits.py`	circuit composition and exact packet-path reconstruction
`dynamics.py`	unitary evolution, leakage bounds, Feshbach reduction, moving frames, canonical connections, Bloch curvature
`channels.py`	Kraus maps, measurements, and packet dephasing
`codes.py`	bit-flip code, syndrome packets, Knill–Laflamme certificate, and recovery
`geometry.py`	cycle and torus spectra, heat trace, spectral dimension, dispersion bounds, and Euclidean volume transfer
`experiments.py`	deterministic numerical certificate suite
`visuals.py`	all code-generated paper figures

The implementation is deliberately finite-dimensional. Infinite-dimensional limits require separate domain, compactness, convergence, positivity, and renormalization arguments and are not inferred from the finite code.

15 Conclusion

The quantum realization of the seed calculus can now be stated precisely. A primitive state is addressed by a declared operator grammar; symmetry projectors split the state and every operator into exact packets; circuit products become sums over packet paths; Hamiltonian symmetry and its controlled breaking are visible as diagonal blocks and leakage; moving packets carry a canonical connection and curvature; channels, measurements, and error-correction conditions admit exact packet formulations; and eliminated sectors return through self-energy and memory rather than disappearing.

The finite theory therefore provides a coherent language for

$$\boxed{\begin{array}{c}\text{seed reachability} + \text{symmetry resolution} + \text{circuit paths} \\ + \text{effective dynamics} + \text{geometric transport} + \text{constraint diagnostics}\end{array}}$$

What it does not yet provide is the physical continuum theorem. The next decisive results would be anomaly-free refinement, uniform effective-operator control, Lorentzian reconstruction, the Einstein infrared limit, and a positive massless spin-two sector. Those requirements are explicit enough to be tested and too substantial to be replaced by finite numerical resemblance.

Finite proof completion

The global projected connection

Paper VII, sixth checked block

Author: Komorebi P. September 6, 2026

The final outstanding finite-sector item, **6.4**, is now checked. The cumulative development contains **323 theorem declarations in 25 Lean modules**. All **19 named finite-sector manuscript items** have formal coverage under the exact hypotheses recorded in the theorem ledger. The five continuum conjectures remain unproved.

The new block contributes **67 theorem declarations across five modules**. These are supporting lemmas and constructions for one remaining manuscript item, not 67 independent manuscript theorems.

New module	Declarations
`GlobalRangeBundle`: actual fibers and smooth local products	22
`LocalProjectedCurvature`: point-local differential identities	6
`GlobalProjectedConnection`: connection laws and gluing	10
`ManifoldProjectedCurvature`: actual global curvature	17
`ProjectedBundleCompletion`: intrinsic and gauge compatibility	12
Total added	67

The completed identity

Let M be a smooth real manifold without boundary, F a complete normed real vector space and $P : M \to \mathrm{End}(F)$ a smooth family of bounded idempotents. For smooth range sections s, define

$$\nabla_X s = P\,\mathrm{d}s[X].$$

The range bundle is constructed from P rather than assumed. Its local connection formulas glue, and the actual covariant commutator satisfies

$$\boxed{(\nabla_X\nabla_Y - \nabla_Y\nabla_X - \nabla_{[X,Y]})s = P\big(\mathrm{d}P[X]\mathrm{d}P[Y] - \mathrm{d}P[Y]\mathrm{d}P[X]\big)P\,s.}$$

The derivative and Lie bracket are Mathlib's actual manifold operators. No global coordinate chart is assumed. The result requires idempotence, not an additional orthogonality hypothesis.

Verification

All 25 modules were freshly compiled in the final run with warnings treated as errors. All 323 declaration-level axiom audits passed. Six constructed objects were separately audited. Their dependencies are confined to `propext`, `Classical.choice` and `Quot.sound`; one upstream theorem is axiom-free. Both deliberate invalid controls were rejected. This records a classical projected-connection result, not a claim of a new differential-geometric theorem.

Actual range bundle

Local products constructed from the projector

The fiber over x is the actual fixed subspace

$$\mathcal{E}_x = \{v \in F : P(x)v = v\} = \operatorname{Ran} P(x).$$

The total space is the subset $\mathcal{E} = \{(x, v) : P(x)v = v\} \subseteq M \times F$, with the induced topology. The construction includes actual local homeomorphisms of this space, not merely abstract transition functions.

An invertible intertwiner near each anchor

For idempotents p, q, set

$$T(p, q) = pq + (I - p)(I - q).$$

Lean proves $T(p, p) = I$ and $pT(p, q) = T(p, q)q$. For an anchor $a \in M$, define

$$U_a = \{x : T(P(x), P(a)) \text{ is invertible}\}.$$

The invertible elements form an open subset of the bounded-operator algebra. Consequently U_a is open and contains a. The family (U_a) covers the whole base. Invertibility is *derived locally*, not assumed at every pair of points.

The mutually inverse local maps

Write $T_a(x) = T(P(x), P(a))$. For $x \in U_a$,

$$B_a(x) : \mathcal{E}_a \longrightarrow \mathcal{E}_x, \quad v \longmapsto T_a(x)v,$$

$$A_a(x) : \mathcal{E}_x \longrightarrow \mathcal{E}_a, \quad w \longmapsto P(a)T_a(x)^{-1}w.$$

The intertwining identity and its inverse counterpart show that the displayed targets are correct. Lean proves both inverse laws, continuity, linearity in the fiber and smooth dependence on the base. Thus

$$\mathcal{E}|_{U_a} \;\cong\; U_a \times \mathcal{E}_a$$

is an actual base-preserving local product homeomorphism.

Smooth overlap maps and rank

The coordinate transition is $g_{ab}(x) = A_b(x)B_a(x)$. It is smooth on $U_a \cap U_b$, agrees with the actual change between local homeomorphisms, and satisfies

$$g_{aa} = I, \qquad g_{bc}g_{ab} = g_{ac}.$$

A continuous linear equivalence between $\mathcal{E}_a$ and $\mathcal{E}_x$ also gives local constancy of fiber dimension. No single constant rank across disconnected components is assumed.

Principal checked objects

`RangeBundle.localProduct`, `RangeBundle.fiberEquiv`, `RangeBundle.smooth_range_bundle_atlas` and `BundleCompletion.localProduct_overlap`, all in namespace `KomorebiPaperVII`.

Intrinsic sections and gluing

A connection on the constructed bundle

The global formula is defined using the actual manifold derivative:

$$\nabla_X s(x) = P(x)\, D_M s(x)[X(x)].$$

Here D_M is represented by `mvfderiv`. A second definition, `GlobalConnection.covariant`, has values in the actual fiber:

$$\nabla s(x) : T_x M \longrightarrow \mathcal{E}_x.$$

It is not only an ambient expression asserted to land in the range.

Connection axioms and smoothness

The new proofs establish range preservation, additivity in the section, linearity over functions in the vector-field argument, and the scalar Leibniz identity

$$\nabla_X(fs) = \mathrm{d}f[X]\, s + f \nabla_X s.$$

Smooth P, smooth sections and smooth vector fields give smooth covariant derivatives. Equality of section germs gives equality of covariant derivatives at the point, so the operation is local.

Smooth sections are genuinely intrinsic

Let $x \in U_a$. If $P(y)s(y) = s(y)$ in a neighborhood of x, then

$$s \text{ is smooth at } x \quad \Longleftrightarrow \quad \big(y \mapsto A_a(y)s(y)\big) \text{ is smooth at } x.$$

The range condition is only local. The proof uses the actual inverse local maps and equality of germs. It follows that smoothness in bundle coordinates does not depend on the chosen chart.

The local connection and its change of coordinates

For a coordinate function u valued in $\mathcal{E}_a$, define

$$\nabla_X^{(a)} u = A_a\big(P\, D_M(B_a u)[X]\big).$$

For every overlap point, Lean proves the operator identity

$$\boxed{\nabla_X^{(b)}(g_{ab}u) = g_{ab}\nabla_X^{(a)} u.}$$

This is proved for arbitrary local coordinate functions. It does not assume that u was obtained from a globally smooth section or that a smooth extension has been supplied. Together with the local smoothness criterion, this closes the gluing step left open in the fifth block.

Principal declarations

`GlobalConnection.nabla_leibniz`, `GlobalConnection.smooth_nabla`, `GlobalConnection.local_connections_glue`, `BundleCompletion.smooth_section_iff` and `BundleCompletion.local_connections_gauge`.

Actual manifold curvature

From differentiated projectors to the two-form

The curvature is defined by the covariant commutator, including the *actual manifold Lie bracket*. It is not defined to equal the desired projector expression.

The flat ambient identity is proved in charts

For a smooth vector-valued function s, the proof identifies directional derivatives with the pullback into each extended chart. It then proves

$$D_M s([X,Y]) = D_M(D_M s[Y])[X] - D_M(D_M s[X])[Y].$$

This uses the chart pullback of the actual vector fields and the ordinary differential identity, then transfers the result back to the manifold. The point-local auxiliary module permits use within arbitrary open chart domains.

Projector and section identities are derived

Differentiating $P^2 = P$ in a tangent direction v yields

$$(\mathrm{d}P[v])P + P\mathrm{d}P[v] = \mathrm{d}P[v], \qquad P\mathrm{d}P[v]P = 0.$$

Differentiating $Ps = s$ gives

$$\mathrm{d}s[v] = \mathrm{d}P[v]s + P\mathrm{d}s[v].$$

Both statements are consequences of actual derivative rules. They are not additional assumptions.

Cancellation of the second derivatives

The product rule and the ambient Lie-bracket identity reduce curvature to

$$R^{\nabla}(X,Y)s = P\mathrm{d}P[X]\mathrm{d}s[Y] - P\mathrm{d}P[Y]\mathrm{d}s[X].$$

Substitute the differentiated range-section identity. The terms containing $P\mathrm{d}P[v]P$ vanish, leaving

$$R^{\nabla}(X,Y)s = P\big(\mathrm{d}P[X]\mathrm{d}P[Y] - \mathrm{d}P[Y]\mathrm{d}P[X]\big)Ps.$$

Lean also proves that this depends only on $X(x)$, $Y(x)$ and $s(x)$, not on their extensions. The operator-valued form is alternating, additive and homogeneous, and lands in the actual fiber. Constant projector bundles have zero curvature form.

Principal declaration

`KomorebiPaperVII.ManifoldCurvature.projected_connection_curvature`

The auxiliary definitions `GlobalConnection.curvature` and `GlobalConnection.curvatureForm` are distinct. The theorem proves their equality on smooth range sections.

Exact formal scope

What the completion does and does not assert

Geometry and scalar conventions

The main global curvature theorem uses a smooth real manifold without boundary, modeled on a complete normed real space, and bounded real-linear endomorphisms of a complete normed real space. Finite-dimensional parameter and quantum spaces fall within the corresponding real differential calculus. Complex quantum spaces can be viewed through their underlying real normed spaces; this block does not introduce a separate complex-bundle typeclass wrapper.

The atlas and global connection constructions also support the more general model-with-corners infrastructure where their hypotheses apply. The *curvature completion claim* is specifically for the boundaryless manifold theorem. A boundary/corner curvature generalization is not claimed.

Explicit smooth atlas, not an assumed one

The implementation exposes the actual total space, local product homeomorphisms, linear transitions, smoothness proofs and cocycle laws. It does not register these as a `ContMDiffVectorBundle` instance. This is an interface choice: no range-bundle atlas, local trivialization, overlap identity or curvature conclusion is inserted as a premise.

The smooth-section equivalence and gauge law make the atlas usable intrinsically. The connection is also defined with values in `fiber(Px)`. Six proof-bearing constructed objects, including both local equivalences and the fiber-valued derivative, have separate axiom audits.

Relationship to the preserved manuscript

The original Paper VII PDF and TeX are unchanged. This supplement and the final ledger are the authoritative record of newly checked coverage. Earlier amendments remain in force, including the additional invertibility hypothesis needed for the Feshbach full-resolvent identity. Completion means the *19-item finite formalization inventory under its recorded assumptions*, not verification of every sentence, numerical experiment or continuum claim in the original PDF.

Conjectures remain conjectures

The five continuum targets are unchanged: anomaly-free closure, cutoff-uniform Feshbach control, reflection-positive reconstruction, the Einstein–Hilbert limit, and a spin-two sector. None is proved by completing the finite projected-connection theorem.

No numerical or dependency overclaim

This block does not rerun the original numerical suite, rebuild Papers I–VI or independently recheck all imported Mathlib proofs. The symbolic bundle definitions are mathematical constructions, not benchmarked numerical bundle-coordinate routines.

Cumulative manuscript coverage

All 19 finite-sector items

The table records named items, not the number of supporting Lean declarations. The JSON ledger records the exact scope and principal source mapping for each item.

Item	Named finite-sector result	Status
2.1	Orbit dimension of one quantum seed	Checked
2.2	Quantum seed-rank formula	Checked
2.3	Local orbit and Schmidt rank	Checked
2.4	One seed for the computational basis	Checked
3.1	Cyclic quantum projectors	Checked
3.2	Tensor/polyphase refinement	Checked
4.1	Exact circuit path expansion	Checked
4.2	Packetwise unitarity conservation	Checked
5.1	Exact symmetry packets	Checked
5.2	Commutator leakage bound	Checked
6.1	Moving-frame Hamiltonian	Checked
6.3	Canonical packet connection	Checked
6.4	Projected connection	Checked
7.1	Block-diagonal measurement invariance	Checked
7.2	Finite Kraus channel	Checked
8.1	Finite Knill-Laflamme criterion	Checked
9.1	Exact Feshbach resolvent	Checked
10.1	Finite Euclidean transfer	Checked
11.1	Low-mode second-order bound	Checked

Cumulative source inventory

256 upstream declarations + 67 new declarations = 323 declarations. All 20 upstream proof files remain byte-for-byte unchanged. The new five modules are listed on page 1. Every declaration in the active inventory has a matching compiler axiom report; the verifier rejects missing, duplicate or unexpected audits.

Completion boundary

Finite-sector items: 19/19 checked. Partially checked finite items: 0. The five continuum conjectures are separate from this denominator. Earlier mathematical corrections are preserved in their supplements and the final theorem ledger.

Verification and reproducibility

Final checked release evidence

Pinned runtime

Lean **4.34.0-rc2**, Mathlib revision

`aeb72f1e9981b6d35b426b16dca7759be3e7ba06`.

The portable kit was restored from seven archive parts. Each part and the combined archive passed SHA-256 checks. The kit itself is not duplicated in the source ZIP.

Final results

All 25 active modules returned exit code 0, with warnings treated as errors. All 323 theorem-level axiom audits and all six separate constructed-object audits passed. Both negative controls returned the expected rejection: a false equality and a deliberate admitted-proof placeholder. All 16 Python verifier regression tests passed.

The authoritative report is `verification/final_finite_release_run/SUMMARY.json`. It records commands, source and transitive dependency hashes, compiled-interface hashes, runtime pins, exit codes and axiom lists. The separate construction report is `verification/final_finite_release_run/CONSTRUCTED_OBJECTS_AUDIT.json`.

Failures are retained

Nine of 16 captured development checks failed before corrections. The diagnostics concerned elaboration, tangent-space aliases, import visibility, local differentiability conversions and simplification. Each captured attempt has its source snapshot and raw log in `verification/development_sixth_block`. None of those failed checks is counted as a final proof pass.

Rechecking

From the extracted release, with the portable runtime restored, run:

```
python scripts/verify.py -kit /path/to/kit -out verification/my_run -timeout 240
python scripts/audit_constructions.py -kit /path/to/kit -run verification/my_run
python scripts/test_verifier.py
```

The verifier compiles local dependencies in order and checks source/interface hashes before accepting resumed work. Use `--max-checks3--resume` when a host has short per-command limits. The second script refuses stale source or build artifacts.

Trust and portability

The launcher invokes the ordinary Lean compiler in module mode with one worker. It does not enforce a RAM cap or weaken proof checking. Generated local build interfaces are excluded from the source archive and can be regenerated. Ordinary compilation and axiom audits are not an independent kernel replay of every imported Mathlib theorem.

Preservation and package integrity

`provenance/UPSTREAM_PRESERVATION_sixth_block.json` records unchanged upstream sources and manuscript files. `SHA256SUMS.txt` inventories distributed files. The companion artifact audit checks the ZIP, hashes, source preservation, proof counts and final reports.

HOW TO READ THIS APPENDIX

Three objects, three questions

Reading source: main text, Sections 1, 7, and 9; finite-completion supplement, pp. 5–6.

A quantum calculation becomes easier to follow when its objects remain visible. This appendix follows a state vector, a density matrix, and a retained part of a dynamical system. They may describe the same experiment, but they do not answer the same question.

$\sqrt{2}\,\psi$

1
0
0
1

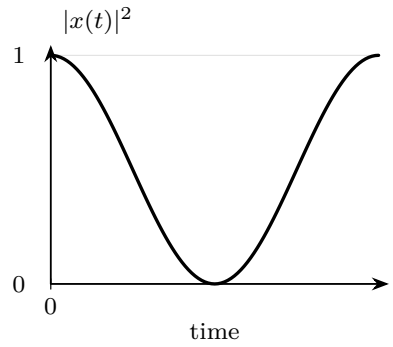
Two amplitudes one pure state

$2\rho = 2\psi\psi^*$

1	0	0	1
0	0	0	0
0	0	0	0
1	0	0	1

Off-diagonal entries retain phase.

Hidden coupling returns.



Amplitudes, not just occupied addresses

For the Bell vector $\psi = (|00\rangle + |11\rangle)/\sqrt{2}$, the two nonzero entries are amplitudes. The density matrix $\rho = \psi\psi^*$ contains additional off-diagonal entries. Those entries matter to measurements that compare the two addresses. Deleting them is a physical channel, not simply rewriting coordinates.

Elimination, not disappearance

When a Hamiltonian couples a retained state to a hidden state, the retained amplitude can leave and later return. Its exact equation then remembers the hidden coupling. A smaller matrix obtained by deleting rows and columns does not generally reproduce that equation.

A route through the paper

Pages V – 3–V – 10 develop seeds, symmetry, and coherent circuit paths. Pages V – 11–V – 15 track leakage and moving subspaces. Pages V – 16–V – 20 distinguish measurement, channels, and recovery. Pages V – 21–V – 27 follow hidden dynamics and finite geometric diagnostics.

The combined reading edition keeps the original 25-page manuscript, then its seven-page v0.6.0 completion supplement, then this appendix. Source lines link to those preserved pages. The appendix is explanatory work: its exact example checks are not new Lean certificates.

> **Read this as:** A packet coordinate change preserves information. A measurement, dephasing operation, or truncation may not. Always identify which operation is actually being performed.

AN ATLAS OF TYPES AND CONVENTIONS

The objects on the page

Main text: Definitions 1.1 and Sections 2–10; supplement, exact formal scope.

The same letter can stand for different kinds of data in different calculations. Keep the following types fixed while reading a formula. In this appendix, vectors are columns, operators compose from right to left, and A^* denotes the conjugate transpose.

Symbol	Object	What it records
$\mathcal{H}, \psi$	Hilbert space and state vector	Addresses and coherent amplitudes.
$\mathcal{A}$	Declared operator algebra	Which linear operations are available.
P_a	Orthogonal packet projector	The whole selected subspace, not one basis vector.
ρ	Positive matrix of trace one	Populations and coherences.
E_k	Positive measurement effect	An outcome probability $\mathrm{Tr}(E_k\rho)$.
K_a	Kraus operator	One branch of a channel; generally not a projector.
H	Self-adjoint Hamiltonian	Unitary real-time evolution $e^{-\mathrm{i}tH}$.
C, ∇	Packet transport generator; connection	Motion of subspaces and differentiation within their ranges.
P, Q	Retained and hidden projectors	A decomposition with $Q = I - P$.
$\Sigma(z)$	Self-energy	Spectral influence of the hidden sector.
$L, T(\tau)$	Nonnegative operator; Euclidean transfer	Decay factors and finite heat-trace diagnostics.

Normalization and indexing

The bit order is the displayed left-to-right tensor order: $X_1 = X \otimes I \otimes I$. A normalized pure state gives $\rho = \psi\psi^*$; a general coefficient matrix used to count a seed orbit need not be normalized. Scaling a nonzero seed does not change its generated linear span.

A Hamiltonian and a Hadamard gate are both traditionally denoted H. Where both appear, the gate is H_{Had} and the Hamiltonian keeps the letter H. Numerical labels on a drawing are coordinates or exact example values, not observations of a physical device.

> **Read this as:** Check the type before interpreting the picture. An idempotent projector, a density matrix, and a channel matrix have different defining equations.

CYCLICITY DEPENDS ON THE GRAMMAR

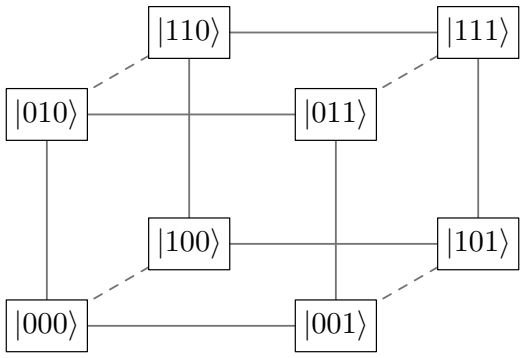
One seed, eight addresses

Main text: Theorem 2.4 and Definition 1.1, pp. 4 and 6.

Start with $\psi_0 = |000\rangle$ and allow the three single-bit flip operators X_1, X_2, X_3. For a binary address $a = (a_1, a_2, a_3)$, put

$$W_a = X_1^{a_1} X_2^{a_2} X_3^{a_3}. \qquad W_a \psi_0 = |a_1 a_2 a_3\rangle.$$

The drawing shows the actual eight states reached by these words. Dashed edges change the first bit between the two displayed squares; the other edges change one of the remaining bits.



Three actual words

$$W_{101} = X_1 X_3$$

$$W_{011} = X_2 X_3$$

$$W_{110} = X_1 X_2$$

$$\dim \operatorname{span}\{W_a |000\rangle\} = 8$$

Two different meanings of “reach”

The eight vectors are orthonormal, so their *linear span* is all of $\mathbb{C}^8$. This is exactly the cyclicity statement for the algebra generated by the flips. It is not a statement that a product of flips alone prepares every normalized vector.

For example, no such word sends $|000\rangle$ to $(|000\rangle + |001\rangle)/\sqrt{2}$. A Hadamard gate, a suitable linear combination in the observable algebra, or a larger preparation grammar is required for that operation. Counting linearly independent addressed vectors and synthesizing a physical gate sequence are separate tasks.

The exact check constructs each 8×8 word matrix, applies it to the seed, and verifies that the matrix of all eight outputs is the identity. It also tests the displayed superposition against every word output. No search over an unspecified gate set is involved.

> **Read this as:** A cyclic seed generates a vector space under an algebra. That conclusion does not automatically provide a unitary preparation procedure for every vector in it.

SCHMIDT RANK AS ACCESSIBLE MULTIPLICITY

Entanglement changes the local span

Main text: Theorem 2.1 and Corollary 2.3, p. 5.

Write a bipartite vector as a coefficient matrix C, with the first subsystem indexing rows. Acting only on that subsystem transforms C to AC. Every resulting row remains in the row space of C, and every independent choice of such rows can be obtained in this way.

$$\dim \operatorname{span}\{(A \otimes I)\psi : A \in M_d(\mathbb{C})\} = d \operatorname{rank} C.$$

Product seed matrix

1	0
0	0

row rank 1
orbit dimension 2

Bell seed matrix $\sqrt{2}C$

1	0
0	1

row rank 2
orbit dimension 4

A acts on the left. It cannot invent a new direction in the row space.

The product vector $|00\rangle$ has matrix $C = \operatorname{diag}(1, 0)$. Its local orbit span contains arbitrary first-subsystem vectors but only the second-subsystem direction $|0\rangle$, so its dimension is two. The Bell vector has $C = I/\sqrt{2}$, rank two, and a four-dimensional local orbit span.

What the formula does not say

The action here allows arbitrary matrices and then takes a linear span. It is not the orbit under local unitary gates alone. Local unitary operations preserve Schmidt coefficients; they do not turn the Bell vector into an arbitrary product vector as a deterministic pure-state transformation.

For the exact calculation, apply the four matrix units E_{ab} to each seed matrix, flatten the outputs, and compute their column rank. The observed dimensions are two and four, respectively. This directly exhibits the rows counted by the theorem rather than inferring them from a drawing of entangled particles.

> **Read this as:** Schmidt rank measures the remote multiplicity directions accessible to the local operator span. It is not permission for unrestricted deterministic local state preparation.

AN EXACT MULTIPLICITY LOWER BOUND

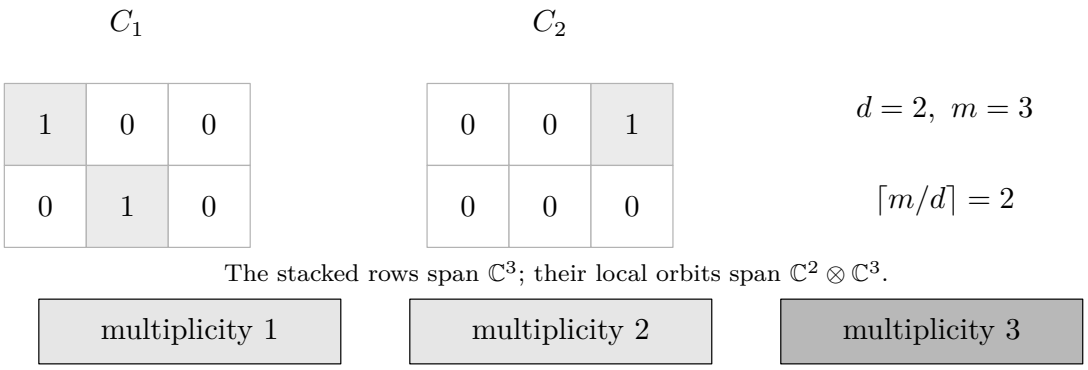
Why one seed is not always enough

Main text: Theorem 2.2, p. 5.

Consider $M_2(\mathbb{C}) \otimes I_3$ acting on $\mathbb{C}^2 \otimes \mathbb{C}^3$. A seed is a 2×3 coefficient matrix. Its two rows span at most two directions in the three-dimensional multiplicity factor. Therefore no single seed is cyclic for this algebra.

$$C_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \qquad C_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

These two matrices provide a complete set of row directions. Their individual orbit dimensions are four and two, and together those spans fill the six-dimensional state space.



From the example to the finite formula

For a block $M_d(\mathbb{C}) \otimes I_m$, k seed matrices contribute at most kd independent multiplicity directions. Hence $k \geq \lceil m/d \rceil$. Partitioning a multiplicity basis into groups of size at most d attains this bound. For a direct sum of such blocks, the worst block determines the required count:

$$\operatorname{srank}_{\mathcal{A}}(\mathcal{H}) = \max_i \lceil m_i/d_i \rceil.$$

This is an invariant of the declared representation of the algebra, not of the Hilbert-space dimension alone. If the available algebra were enlarged to the full matrix algebra on $\mathbb{C}^6$, every nonzero seed would instead be cyclic.

The example checks both the ranks of the separate coefficient matrices and the rank three of their stacked row matrix. The argument explains why the construction meets the lower bound; a successful example alone would not prove minimality.

> **Read this as:** Missing multiplicity directions cannot be created by reusing the same left action. Additional seeds or a larger operator algebra are genuinely different resources.

CHARACTER PROJECTORS YOU CAN MULTIPLY

Four phases resolve a cyclic shift

Main text: Theorem 3.1, p. 6.

On four sites, let $S|k\rangle = |k+1 \bmod 4\rangle$ and let $\omega = \mathrm{i}$. The character projectors are the exact matrices

$$P_j = \frac{1}{4}\sum_{r=0}^{3} \mathrm{i}^{-jr} S^r, \qquad v_j(k) = \frac{1}{2}\mathrm{i}^{-jk}, \qquad P_j = v_j v_j^*.$$

All entries belong to the Gaussian rationals. Thus no approximation to a root of unity is needed in this model.

$2V$, with $V^*V = I$

	$j=0$	$j=1$	$j=2$	$j=3$
site 0	1	1	1	1
site 1	1	$-i$	-1	i
site 2	1	-1	1	-1
site 3	1	i	-1	$-i$

$$Sv_0 = 1v_0 \qquad P_0 = v_0 v_0^*$$

$$Sv_1 = iv_1 \qquad P_1 = v_1 v_1^*$$

$$Sv_2 = -1v_2 \qquad P_2 = v_2 v_2^*$$

$$Sv_3 = -iv_3 \qquad P_3 = v_3 v_3^*$$

Each column has four entries of magnitude 1/2.

Reading the columns

Each column v_j is a normalized wave around the four sites. Advancing the address multiplies it by i^j. The entries show phase differences rather than four different probability distributions: every column has uniform site probability 1/4.

Multiplying the actual projectors gives $P_j P_k = 0$ for $j \neq k$, $P_j^2 = P_j$, and $\sum_j P_j = I$. In particular, the complete packet tuple $(P_j\psi)_j$ retains the original vector exactly. Recording only the four packet norms is a different operation and can discard phase information between packets.

The four-site shift will return on V–27. There it diagonalizes the Hamiltonian $L = 2I - S - S^*$, and its four character phases explain perfect transfer between opposite sites. The same fixed projectors serve as coordinates, symmetry sectors, and a complete interference ledger.

> **Read this as:** A character projector retains a complex subspace component. Its norm is only one scalar summary of that component, not a substitute for it.

TENSOR REFINEMENT BY TOTAL PARITY

A coarse packet can still contain coherence

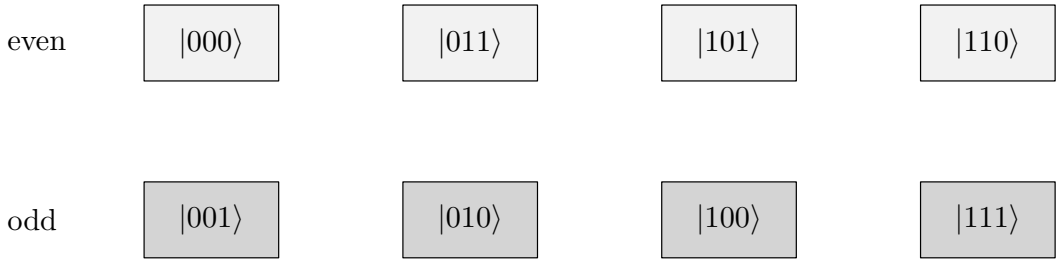
Main text: Theorem 3.2 and Numerical Certificate 3.3, p. 7.

For three qubits, the eight fine projectors are $P_a = |a\rangle\langle a|$. Grouping addresses by Hamming-weight parity gives

$$P_{\text{even}} = P_{000} + P_{011} + P_{101} + P_{110} = \frac{I + Z \otimes Z \otimes Z}{2}, \qquad P_{\text{odd}} = I - P_{\text{even}}.$$

This is a sum of subspaces, not an averaging of their coordinates. The rank-four even packet retains all complex amplitudes on its four children.

Each coarse projector has rank four; all eight fine projectors have rank one.



Refinement is not measurement

The fine packet description and the coarse packet description each reconstruct the full vector when every component is retained. Replacing a state by the mixture $\sum_a P_a \rho P_a$ is instead a dephasing channel. That operation removes cross-child coherence and need not agree with dephasing only between even and odd sectors.

For instance, a superposition of $|000\rangle$ and $|011\rangle$ lies entirely in the even packet. Its internal relative phase survives a parity measurement but is lost by a complete computational-basis measurement if outcomes are forgotten. The two measurements use projectors from the same refinement hierarchy but expose different information.

The exact check verifies all eight addresses, their parity grouping, the rank of both coarse projectors, and equality with the tensor expression above. The general residue version replaces parity by total label modulo N and groups the tensor descendants by their summed residue.

> **Read this as:** A coarse packet is a whole subspace. Internal coherence survives until an operation actually discards it; changing the address description alone does not do so.

INTERMEDIATE PACKETS ARE NOT FINAL PACKETS

Track the Bell state through the circuit

Main text: Theorem 4.1 and Example 4.3, p. 8.

The input is $|00\rangle$. First apply a Hadamard to the first qubit, then a controlled flip:

$$|00\rangle \xmapsto{H_{\mathrm{Had}}\otimes I} \frac{|00\rangle + |10\rangle}{\sqrt{2}} \xmapsto{\mathrm{CNOT}} \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

The displayed columns keep amplitudes, including their normalization, at every step. The middle vector occupies both parity packets. The final vector occupies only the even packet.

	seed		after $H \otimes I$		after CNOT
00	1	00	$\frac{\sqrt{2}}{2}$	00	$\frac{\sqrt{2}}{2}$
01	0	01	0	01	0
10	0	10	$\frac{\sqrt{2}}{2}$	10	0
11	0	11	0	11	$\frac{\sqrt{2}}{2}$
	even only		both parity packets		even only, with coherence

How the exact path expansion works

Insert $P_{\mathrm{even}} + P_{\mathrm{odd}} = I$ between the two gates. Each summand records a particular intermediate packet, while the boundary projectors select its input and output packets. Summing all terms recovers the original 4×4 circuit matrix:

$$\sum_{a,b,c} P_b \,\mathrm{CNOT}\, P_a (H_{\mathrm{Had}} \otimes I) P_c = \mathrm{CNOT}(H_{\mathrm{Had}} \otimes I).$$

This expansion is an identity even though the intermediate parity is not measured. Physically inserting a measurement would alter the problem by removing interference between its alternatives. A coordinate resolution in a derivation should not be mistaken for an experimental intervention.

In the finite check, all eight parity-indexed matrix terms are constructed and summed. The matrix equality is checked before specializing to the input vector. It therefore verifies the decomposition for every input of this example, not merely agreement on the Bell seed.

> **Read this as:** A final invariant packet does not imply that each preceding gate preserved it. Follow the amplitudes through the actual intermediate resolutions.

A COMPLETE DESTRUCTIVE-INTERFERENCE LEDGER

Why path probabilities do not add

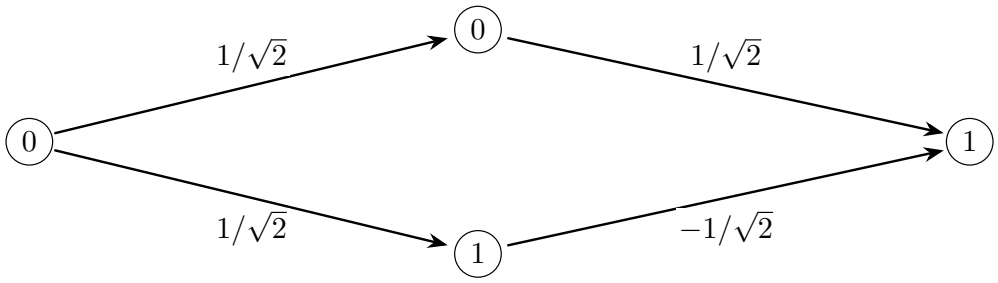
Main text: Theorems 4.1–4.2, p. 8.

Two consecutive Hadamard gates multiply to the identity. Starting from $|0\rangle$, there are two resolved paths to output $|1\rangle$. Their signed products are

$$\langle 1|H_{\mathrm{Had}}|0\rangle\langle 0|H_{\mathrm{Had}}|0\rangle = \frac{1}{2}, \qquad \langle 1|H_{\mathrm{Had}}|1\rangle\langle 1|H_{\mathrm{Had}}|0\rangle = -\frac{1}{2}.$$

Their sum is zero. Squaring each separately gives a different answer.

Two actual Hadamard paths, not classical alternatives



$$\text{amplitude} = 1/2 - 1/2 = 0 \quad \neq \quad \text{sum of path probabilities} = 1/2$$

Where the missing term goes

For path amplitudes a, b, the outcome probability is $|a+b|^2 = |a|^2 + |b|^2 + 2\Re(a\bar{b})$. Here the last term is $-1/2$, canceling the first two terms. The path picture retains this information only if its edges carry complex amplitudes rather than unsigned reachability or probabilities.

The packetwise unitarity identity is likewise an operator identity, including the off-diagonal cancellations:

$$\sum_b U_{ba}^* U_{bc} = \delta_{ac} P_a, \qquad U_{ba} = P_b U P_a.$$

It cannot be replaced by the assertion that every path individually has unit norm. The exact checks verify all four choices of a, c for the two rank-one packets, as well as the two opposite-output path products.

A measurement at the intermediate stage would remove the cross term. In that modified experiment the two alternatives really would contribute total probability $1/2$ to output one. Thus the false calculation accidentally describes a different experiment.

> **Read this as:** Insert projectors algebraically to resolve paths, then add amplitudes. Add probabilities only when the physical setup has actually made the alternatives incoherent.

ONE HAMILTONIAN IN TWO ORDERS

A symmetry reveals blocks, not approximations

Main text: Theorem 5.1 and the two-qubit example, pp. 8–10.

Take the exact rational Hamiltonian

$$H = X \otimes X + \frac{3}{8}(Z \otimes I + I \otimes Z).$$

It commutes with total parity $Z \otimes Z$. Reordering the computational basis into even addresses $(00, 11)$ and odd addresses $(01, 10)$ displays the invariant subspaces explicitly.

Computational order

	00	01	10	11
00	$\frac{3}{4}$	0	0	1
01	0	0	1	0
10	0	1	0	0
11	1	0	0	$-\frac{3}{4}$

same operator
different order

Parity order

	00	11	01	10
00	$\frac{3}{4}$	1	0	0
11	1	$-\frac{3}{4}$	0	0
01	0	0	0	1
10	0	0	1	0

Even eigenvalues: $\pm 5/4$. Odd eigenvalues: ± 1.

$$H_{\text{even}} = \begin{pmatrix} 3/4 & 1 \\ 1 & -3/4 \end{pmatrix}, \qquad H_{\text{odd}} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The even block squares to $(25/16)I$, so its eigenvalues are $\pm 5/4$. The odd block squares to I, so its eigenvalues are ± 1. Exponentiating either block gives the corresponding part of the exact evolution; the off-packet transfer stays zero for every real time.

What has changed?

Only the order of basis vectors has changed. No matrix entries, amplitudes, or energy levels were discarded. This is why the decomposition is exact. Keeping just the even block would be a separate restriction of the state space, justified for initial data known to be even and for dynamics that preserve it.

The next page breaks a symmetry in a smaller model. There the cross-packet entries no longer vanish, and their effect must be controlled rather than hidden by reordering. The exact tests here multiply the commutator, verify the displayed permutation, and compute all four eigenvalues symbolically.

> **Read this as:** Block diagonalization uses an invariant decomposition already present in the operator. It is not evidence that an arbitrary chosen packet is invariant.

COMMUTATORS AND PROBABILITY BOUNDS

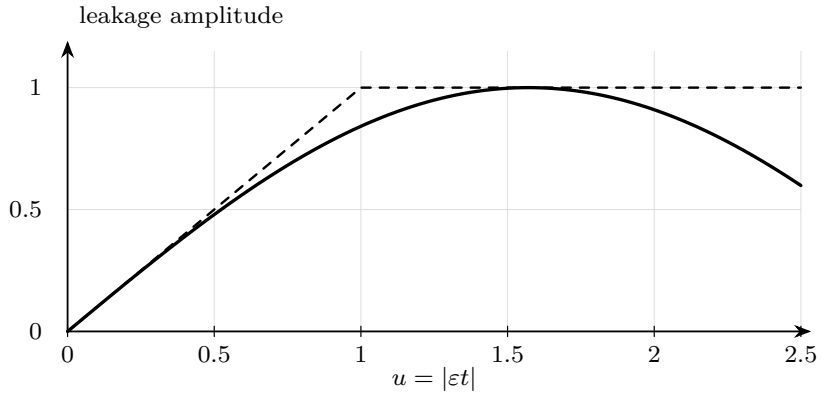
Quantify the amplitude that escapes

Main text: Theorem 5.2, p. 9.

For a two-level system, choose $H = \varepsilon X$, $P = |0\rangle\langle 0|$, and $Q = I - P$. Direct exponentiation gives

$$e^{-\mathrm{i}tH} = \cos(\varepsilon t)I - \mathrm{i}\sin(\varepsilon t)X, \quad \left\|Qe^{-\mathrm{i}tH}P\right\| = |\sin(\varepsilon t)|, \quad \|[H,P]\| = |\varepsilon|.$$

The general commutator estimate becomes the elementary bound $|\sin u| \le |u|$. Unitarity also bounds the amplitude by one.



$H = \varepsilon X$

solid: $|\sin u|$
dashed: $\min(1, u)$

Probability is $\sin^2 u$,
not $|\sin u|$.

From a norm to a probability

For this initial state, the escape probability is $\sin^2(\varepsilon t)$, not $|\sin(\varepsilon t)|$. Consequently

$$\Pr(\text{hidden packet at } t) \le \min\{1, \varepsilon^2 t^2\}.$$

At $\varepsilon t = \pi/6$, the amplitude is $1/2$ and the probability is $1/4$. Near zero, amplitudes are first order while escape probabilities are second order. Mixing those two quantities produces an incorrect interpretation of a leakage plot.

The figure samples closed-form functions for display. The supporting check verifies the exact exponential identity, its unitarity, the squared leakage matrix, and the special-time value. The bound itself is supplied by the paper's theorem, not established by sampling the curve.

At larger times amplitude can return from the hidden packet. Thus the commutator bound is a worst-case guarantee, not a prediction that leakage rises monotonically forever.

> **Read this as:** The norm estimate controls amplitude. Square it for a normalized-state probability bound, and retain the distinction between a bound and the actual oscillatory dynamics.

NOT ALL MOTION IS PHYSICAL EVOLUTION

A moving frame creates a connection term

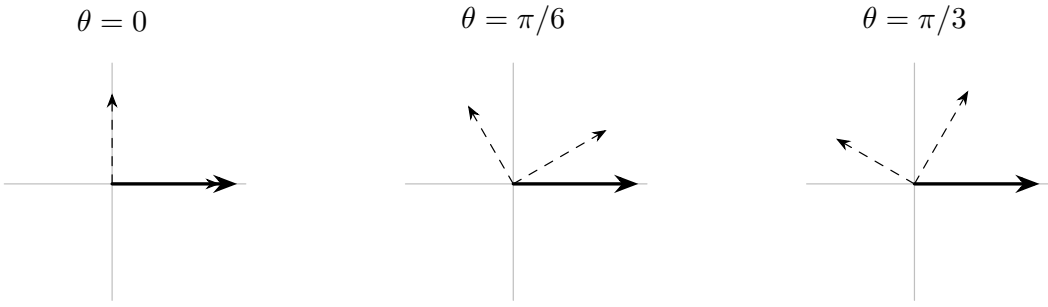
Main text: Theorem 6.1, p. 11.

Set the laboratory Hamiltonian to zero and keep $\psi(t) = |0\rangle$ fixed. Describe it in rotating coordinates,

$$U(t) = \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}, \qquad y(t) = U(t)^* \psi(t) = \begin{pmatrix} \cos t \\ -\sin t \end{pmatrix}.$$

The vector has not moved in the laboratory, but its coordinate entries have changed.

Solid vector: fixed in the laboratory. Dashed axes: rotating frame.



$(1, 0)$ $(\sqrt{3}/2, -1/2)$ $(1/2, -\sqrt{3}/2)$

The missing term can be computed exactly

Let $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Since $U^*\dot{U} = J$, the coordinate Hamiltonian is

$$H_{\text{frame}} = -\mathrm{i}J = \begin{pmatrix} 0 & \mathrm{i} \\ -\mathrm{i} & 0 \end{pmatrix}, \qquad \mathrm{i}\dot{y} = H_{\text{frame}} y.$$

This matrix is Hermitian. Omitting it would predict a constant y, contradicting the displayed coordinates. In this example the norm of the omitted time derivative is exactly one, so the failure is not a small numerical residual.

For a nonzero laboratory Hamiltonian, the same product rule adds the transformed physical term U^*HU. Thus $U^*HU - \mathrm{i}U^*\dot{U}$ separates the physical generator from the coordinate correction. Both pieces belong to the same equation but have different origins.

The drawing shows actual rotated axes at three angles, not three different laboratory trajectories. The check differentiates the exact matrices and verifies the coordinate Schrödinger equation symbolically. A finite-difference test is unnecessary for this model.

> **Read this as:** When coordinates move, even a stationary physical vector acquires changing components. The connection term accounts for that change; it is not an optional force added to the laboratory system.

THE CANONICAL PACKET CONNECTION

Differentiate the subspace itself

Main text: Theorem 6.3, p. 12.

Continue with $P(t) = U(t)\,\mathrm{diag}(1,0)U(t)^*$ and $Q(t) = I - P(t)$. Differentiating $P^2 = P$ gives

$$\dot{P}P + P\dot{P} = \dot{P}, \qquad P\dot{P}P = 0.$$

The second identity says that the derivative has no component entirely inside the current packet. It moves the subspace relative to its complement.

$P(0)$

1	0
0	0

$\dot{P}(0)$

0	1
1	0

C

0	-1
1	0

$$P\dot{P}P = 0, \qquad C^* = -C, \qquad [C, P] = \dot{P}.$$

For this complete pair of packets, define

$$C = \dot{P}P + \dot{Q}Q.$$

Exact calculation yields the constant matrix $C = J$, with $C^* = -C$ and $[C, P] = \dot{P}$. At $t = 0$, the matrices in the plate let the reader verify these identities by multiplying four small arrays.

From two packets to a complete family

For differentiable orthogonal projectors P_a summing to the identity, the paper constructs

$$C = \sum_a \dot{P}_a P_a = \frac{1}{2}\sum_a [\dot{P}_a, P_a].$$

It transports every packet derivative simultaneously. No basis within a packet is needed for this formula. A basis choice would add internal coordinate data without changing which subspaces are selected.

The completeness of the family matters in deriving skew-Hermiticity and the simultaneous transport identities. Using only one summand of an incomplete list is not automatically the same connection. The example retains both P and Q throughout, even when the picture emphasizes only the first line.

> **Read this as:** The projector derivative describes motion between subspaces. The canonical connection reconstructs that motion from the complete family without choosing internal packet bases.

THE RANGE BUNDLE IS CONSTRUCTED LOCALLY

A chart failure is not a singular fiber

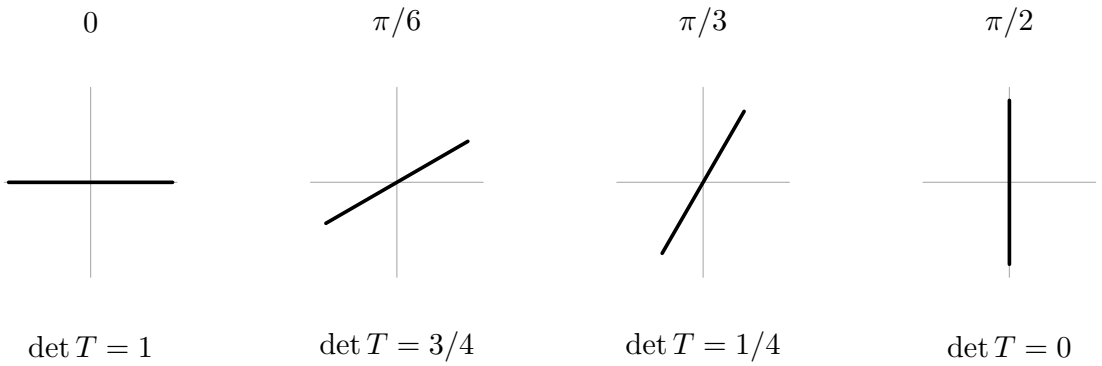
Reading source: finite-completion supplement, pp. 2–4; main text, Classical Input 6.4.

The fiber of a projector field is the actual fixed subspace $E_x = \{v : P(x)v = v\}$. To compare it with a nearby anchor fiber E_a, the completion proof constructs

$$T_a(x) = P(x)P(a) + (I - P(x))(I - P(a)).$$

It equals the identity at the anchor and intertwines the two projectors. Where it is invertible, its restriction provides local fiber coordinates.

Ranges of $P(\theta)$; the anchor is the horizontal line.



At the last position, use a new anchor. The fiber still has dimension one.

An exact example of the domain restriction

For the rotating line and anchor $P(0)$,

$$T_0(t) = \cos t\, U(t), \qquad \det T_0(t) = \cos^2 t.$$

The chart fails at $t = \pi/2$, but $P(\pi/2) = \mathrm{diag}(0, 1)$ is a perfectly regular rank-one projector. Choosing a vertical anchor supplies a new chart there. What failed was the comparison with one reference fiber, not the fiber itself.

In general, the open sets where $T_a(x)$ is invertible cover the base. The inverse chart maps and smooth overlap maps are constructed from the actual matrices. Their cocycle identities express consistent coordinate changes, rather than assuming a global basis exists.

The v0.6.0 supplement proves the intrinsic connection and its curvature using actual manifold derivatives. Its scope includes bounded idempotents, not only orthogonal ones. The geometric illustration here is finite-dimensional and orthogonal for transparency; that convenience is not promoted to an extra premise of the recorded theorem.

> **Read this as:** A smooth bundle is built from compatible local charts. The singularity of one chart determinant can be harmless, provided another anchor covers the same smooth fiber.

AN EXACT BLOCH-PROJECTOR CALCULATION

Curvature measures a local area response

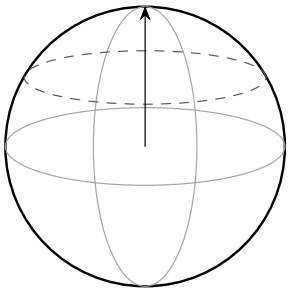
Main text: Section 6.3, pp. 12–13; completion supplement, curvature theorem.

On the unit sphere, set $n = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$ and

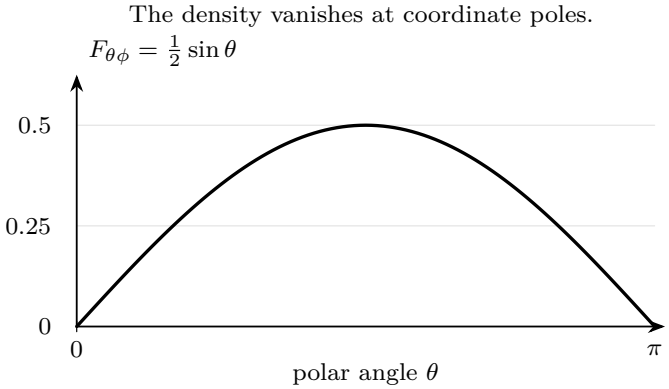
$$P(\theta,\phi) = \frac{1}{2}(I + n\cdot\sigma).$$

This is a smooth rank-one orthogonal projector. The intrinsic curvature is the endomorphism-valued two-form $P(\mathrm{d}P\wedge\mathrm{d}P)P$. With the paper's scalar convention, its component is

$$F_{\theta\phi} = -\mathrm{i}\,\mathrm{Tr}(P[\partial_\theta P, \partial_\phi P]) = \frac{1}{2}\sin\theta.$$



Cap flux at $\theta = \pi/3$ is $\pi/2$.



A local frame gives the same answer

Away from the south pole, use $u_N = (\cos(\theta/2), e^{\mathrm{i}\phi}\sin(\theta/2))^T$. Then $P = u_N u_N^*$ and the scalar one-form is

$$A_N = -\mathrm{i}u_N^*\mathrm{d}u_N = \frac{1-\cos\theta}{2}\,\mathrm{d}\phi, \qquad \mathrm{d}A_N = \frac{1}{2}\sin\theta\,\mathrm{d}\theta\wedge\mathrm{d}\phi.$$

Thus the curvature flux through the cap $0 \le \theta \le \theta_0$ is $\pi(1-\cos\theta_0)$. At $\theta_0 = \pi/3$, it is exactly $\pi/2$. The plate's latitude is an illustration of this cap; the formula, not its apparent area on the page, supplies the value.

The vanishing of the coordinate density at a pole does not mean the geometric curvature disappears there. The spherical area form also carries the factor $\sin\theta$. A phase assigned to parallel transport additionally requires an orientation and a connection-sign convention; flux and phase should not be interchanged without those choices.

> **Read this as:** The projector formula and a local frame calculation agree. Curvature is intrinsic even when no single smooth frame covers the whole sphere.

THE ANSWER DEPENDS ON ITS EFFECTS

Which coherence can a measurement see?

Main text: Theorem 7.1, p. 14.

For the Bell density matrix $\rho = |\Phi^+\rangle\langle\Phi^+|$, parity dephasing does nothing: both nonzero amplitudes already lie in the even subspace. Computational-basis dephasing instead removes the entries between addresses 00 and 11.

2ρ; also $2D_{\text{parity}}(\rho)$

	00	01	10	11
00	1	0	0	1
01	0	0	0	0
10	0	0	0	0
11	1	0	0	1

same diagonal
different coherence

$2D_{\text{computational}}(\rho)$

	00	01	10	11
00	1	0	0	0
01	0	0	0	0
10	0	0	0	0
11	0	0	0	1

Bell-projector outcome: 1 on the left, 1/2 on the right.

Every computational-basis effect $E_a = |a\rangle\langle a|$ has the same expectation on both matrices, namely 1/2 at 00 and 11 and zero elsewhere. These measurements read only the diagonal. They cannot distinguish the pure Bell state from this incoherent mixture.

A measurement that resolves the difference

Use the valid two-outcome measurement with effects $E = \rho$ and $I - E$. Then

$$\operatorname{Tr}(E\rho) = 1, \qquad \operatorname{Tr}(ED_{\text{computational}}(\rho)) = \frac{1}{2}.$$

The effect itself is not diagonal in the fine computational packets. It compares their coherent combination, so the paper's block-diagonal invariance hypothesis does not apply.

This example contains three operations that should not be conflated: representing a state by all of its packet matrix blocks, applying a dephasing channel, and reading one specified measurement. Only the second discards the off-diagonal blocks. The third may or may not notice that loss, depending on its effects.

The exact checks construct all four fine projectors, both parity projectors, both dephased states, and the outcome traces. They reject the stronger statement that *every* measurement is invariant under dephasing.

> **Read this as:** A measurement's inability to see a coherence does not make that coherence absent. It states a restriction on the observable, not an equality of the underlying states.

POPULATIONS AND COHERENCES TRANSFORM DIFFERENTLY

A channel acts on four matrix entries

Main text: Classical Input 7.2 and packet superoperators, p. 14.

Choose amplitude damping with $\gamma = 9/25$:

$$K_0 = \begin{pmatrix} 1 & 0 \\ 0 & 4/5 \end{pmatrix}, \qquad K_1 = \begin{pmatrix} 0 & 3/5 \\ 0 & 0 \end{pmatrix}, \qquad K_0^* K_0 + K_1^* K_1 = I.$$

The channel $\Phi(\rho) = \sum_a K_a \rho K_a^*$ is trace preserving and completely positive. The special parameter keeps every displayed coefficient rational.

Superoperator on matrix entries

	00	01	10	11
00	1	0	0	$\frac{9}{25}$
01	0	$\frac{4}{5}$	0	0
10	0	0	$\frac{4}{5}$	0
11	0	0	0	$\frac{16}{25}$

$\gamma = 9/25$

coherence factor $4/5$

Output from $|+\rangle$

$\frac{17}{25}$	$\frac{2}{5}$
$\frac{2}{5}$	$\frac{8}{25}$

$$\operatorname{Tr}\rho = 1, \quad \det\rho = 36/625 > 0$$

The superoperator uses the entry order $(\rho_{00}, \rho_{01}, \rho_{10}, \rho_{11})$. Population leaves the excited state with factor $9/25$, while coherences contract by $4/5$. These are not the same rate or the same coordinate.

For $\rho_+ = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, the exact output is

$$\Phi(\rho_+) = \begin{pmatrix} 17/25 & 2/5 \\ 2/5 & 8/25 \end{pmatrix}.$$

Its trace is one and determinant is $36/625 > 0$, so it is a mixed positive state. Keeping only population-transfer arrows would not specify its off-diagonal entries.

The complete-positivity certificate

The Choi matrix is an explicit sum of two outer products of vectorized Kraus operators. Its eigenvalues are $41/25, 9/25, 0, 0$, all nonnegative. This gives an independent finite certificate in addition to Kraus normalization. The Choi matrix has trace two under the unnormalized maximally entangled convention used by the check.

> **Read this as:** A packet channel acts on pairs of packet addresses, because density matrices have two indices. Population transfer alone does not reconstruct a quantum channel.

WHY COMPLETE POSITIVITY IS THE RELEVANT CONDITION

An ancilla tests more than positivity

Main text: Classical Input 7.2, the arbitrary-auxiliary-dimension clause, p. 14.

Ordinary transpose T sends a positive single-system matrix to another positive matrix and preserves its trace. Those properties are insufficient for a quantum channel on a subsystem of an entangled state. The test must include $\mathrm{id}\otimes\mathsf{T}$ on the larger system.

$2(\mathrm{id}\otimes\mathsf{T})(\rho_{\mathrm{Bell}})$

00	1	0	0	0
01	0	0	1	0
10	0	1	0	0
11	0	0	0	1

$$w = (|01\rangle - |10\rangle)/\sqrt{2}$$

$$w^*(\mathrm{id}\otimes\mathsf{T})(\rho)w = -1/2$$

Transpose is positive on one system.
It is not completely positive.

Apply this extended map to the Bell density matrix. The result is

$$(\mathrm{id}\otimes\mathsf{T})(\rho_{\mathrm{Bell}}) = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Its middle block has one negative eigenvalue. More concretely, the normalized vector $w = (|01\rangle - |10\rangle)/\sqrt{2}$ satisfies

$$w^*(\mathrm{id}\otimes\mathsf{T})(\rho_{\mathrm{Bell}})w = -\frac{1}{2}.$$

This single exact quadratic-form value certifies failure of positivity on the composite system. No numerical eigenvalue tolerance is needed.

Why the Kraus form avoids this problem

For every auxiliary dimension and positive X, each term $(I \otimes K_a)X(I \otimes K_a)^*$ is positive. Their sum is positive as well. This argument explains the word *completely*; checking one input density matrix in the original dimension is not an equivalent test.

The transpose counterexample is an explanatory finite calculation, not a new addition to the formal theorem count. It makes visible the hypothesis that the manuscript's Kraus result already supplies. The exact check constructs the partial transpose with the stated tensor indexing and evaluates its negative witness directly.

> **Read this as:** A map can look harmless on an isolated system and fail on an entangled extension. Complete positivity retains compatibility with every auxiliary system.

FOUR EXACT SYNDROME PACKETS

Find the error without reading the logical state

Main text: Classical Input 8.1 and the bit-flip example, pp. 15–16.

Encode a qubit by $|0_L\rangle = |000\rangle$ and $|1_L\rangle = |111\rangle$. Let P project onto this code and take errors $E_a \in \{I, X_1, X_2, X_3\}$. Their images are four orthogonal rank-two packets $Q_a = E_a P E_a^*$ that sum to the full identity.

Channel probabilities: 1/10, 2/10, 3/10, 4/10.

no error	X_1	X_2	X_3
$\|000\rangle$	$\|100\rangle$	$\|010\rangle$	$\|001\rangle$
$\|111\rangle$	$\|011\rangle$	$\|101\rangle$	$\|110\rangle$
ratio 3 : 4	ratio 3 : 4	ratio 3 : 4	ratio 3 : 4

Use the logical state $\psi = (3|0_L\rangle + 4|1_L\rangle)/5$ and channel probabilities $w = (1, 2, 3, 4)/10$. The noisy state is $\sum_a w_a E_a \psi\psi^* E_a^*$. A syndrome measurement returns label a with probability w_a, independent of the logical amplitudes.

The recovery is an actual channel

The operators $R_a = E_a^* Q_a$ satisfy $\sum_a R_a^* R_a = I$. Applying the recovery $\mathcal{R}(X) = \sum_a R_a X R_a^*$ to the noisy state gives exactly $\psi\psi^*$. The off-diagonal logical coherence 12/25 is recovered as well as the populations 9/25 and 16/25.

The Knill–Laflamme compressions are $P E_a^* E_b P = \delta_{ab} P$. Their scalar form says that the error labels do not resolve the encoded vector. The exact checks evaluate all sixteen compressions, all sixteen syndrome products, the four probabilities, the recovery normalization, and the full recovered matrix.

This code corrects the specified single-bit flips. It is not claimed to correct arbitrary errors on three qubits, or even arbitrary single-qubit errors. The next example exhibits an error outside its protected family.

> **Read this as:** A syndrome identifies an error subspace while leaving a two-dimensional logical subspace intact. Measuring the two logical basis states separately would destroy the information the recovery must preserve.

THE SCALAR COMPRESSION TEST MATTERS

A harmless-looking syndrome can hide a logical error

Main text: Classical Input 8.1, pp. 15–16; normalized recovery supplements.

Apply $Z_1 = Z \otimes I \otimes I$ to the same repetition code. It sends $|000\rangle$ to itself and $|111\rangle$ to its negative. Both states remain inside the original code, so the previous syndrome partition reports no bit flip.

$$\frac{3|000\rangle + 4|111\rangle}{5} \longmapsto \frac{3|000\rangle - 4|111\rangle}{5}.$$

The populations are unchanged, but the relative phase is not.

$V^* Z_1 V$

1	0
0	-1

before: $(3, 4)/5$

after: $(3, -4)/5$

same code support
different relative phase

Required scalar form cI

c	0
0	c

A trace of zero does not make $\mathrm{diag}(1, -1)$ the zero operator.

Let V be the code isometry with columns $|000\rangle, |111\rangle$. For the proposed error family $\{I, Z_1\}$, its off-diagonal error compression is

$$V^* I^* Z_1 V = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

This matrix is not cI for any scalar c. Thus the Knill–Laflamme condition fails for that family. Its trace is zero, but its action on the code is certainly nonzero. A trace test alone would miss the obstruction.

Known correction versus family recovery

If it is known with certainty that Z_1 occurred, applying Z_1 again reverses it. The failed criterion concerns a *single recovery valid for the whole declared unknown-error family*, not invertibility of each error considered separately. This distinction prevents a misleading conclusion that no phase error can ever be undone.

The finite calculation checks the actual compressed matrix and rejects its equality to zero. The scalar obstruction is then transparent from the two unequal diagonal entries.

> **Read this as:** Correction must preserve all logical superpositions for the declared error family. Staying in the code and preserving basis probabilities are not sufficient.

TWO INVERTIBILITY CONDITIONS, NOT ONE

Hidden states return as spectral dependence

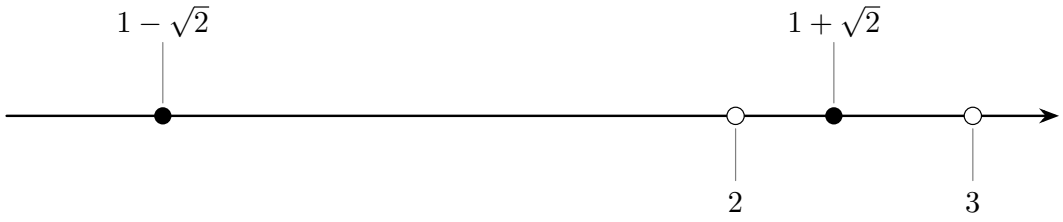
Main text: Theorem 9.1, pp. 16–17; first-block amendments and v0.6.0 ledger.

Take $H = \begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix}$ and retain its first coordinate. The eliminated block is $H_{QQ} = 2$. Whenever $z - 2$ and the full $zI - H$ are invertible,

$$H_{\mathrm{eff}}(z) = \frac{1}{z-2}, \qquad \left[(zI - H)^{-1}\right]_{00} = \frac{1}{z - 1/(z-2)} = \frac{z-2}{z^2 - 2z - 1}.$$

This is the corrected hypothesis set recorded in the formal release.

Filled: full resolvent poles. Open at 2: only the eliminated inverse fails.



At $z = 3$: exact retained resolvent $1/2$, bare truncation $1/3$.

At $z = 3$, the exact retained resolvent is $1/2$. Bare truncation uses only $H_{PP} = 0$ and gives $1/3$, an error of $1/6$. The omitted coupling has changed the response even though the retained diagonal entry did not change.

A pole can belong to the elimination chart

At $z = 2$, the eliminated inverse does not exist, so the displayed Schur formula cannot be applied there as written. Nevertheless, the full resolvent exists and its retained entry is zero. The rational expression has a continuation to this value. By contrast, $1 \pm \sqrt{2}$ are genuine full-resolvent poles.

The full-invertibility hypothesis is independently necessary. At $z = 0$, $H = \mathrm{diag}(0, -1)$ has an invertible eliminated block in $zI - H$, but the full matrix is singular. The original prose omitted this condition; the preserved amendments restore it. This appendix uses the corrected form throughout.

> **Read this as:** An exact elimination formula has a spectral domain. Check both inverses before evaluating it, and distinguish poles of an intermediate inverse from poles of the full physical resolvent.

AN EXACT QUANTUM MEMORY EQUATION

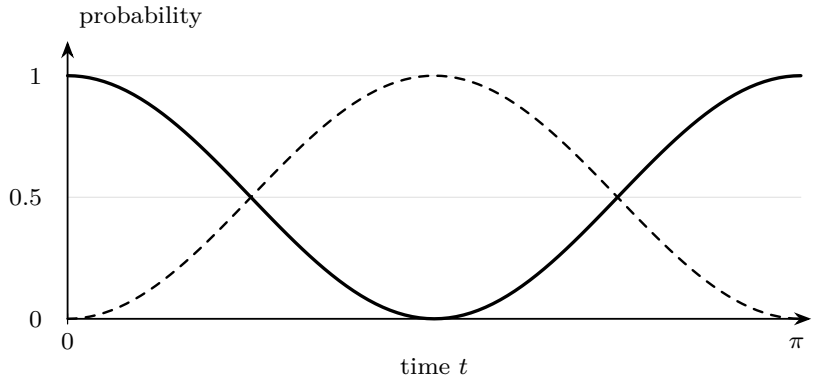
The smaller system remembers the discarded one

Main text: Section 9.2, pp. 17–18.

Choose $H = X$, initial state $(1, 0)^T$, and retain the first coordinate. The coupled equations and their solution are

$$\mathrm{i}\dot{x} = y, \qquad \mathrm{i}\dot{y} = x, \qquad x(t) = \cos t, \quad y(t) = -\mathrm{i}\sin t.$$

The hidden probability returns periodically. Deleting the hidden row and column would instead leave the zero Hamiltonian and predict $x(t) = 1$.



solid: retained
dashed: hidden

$K(t) = 1$

The two probabilities sum to one.

Eliminate exactly, not approximately

Integrating the second equation with $y(0) = 0$ gives $y(t) = -\mathrm{i}\int_0^t x(s)\,ds$. Substitution produces

$$\dot{x}(t) = -\int_0^t x(s)\,ds.$$

The memory kernel is the constant function one. Differentiating once more recovers $\ddot{x} = -x$ with $x(0) = 1$ and $\dot{x}(0) = 0$, explaining the cosine solution. A nonzero hidden initial state would add its own forcing term; it cannot be silently ignored.

The full propagator is a unitary group, but its retained scalar entry is not a semigroup. At $t = s = \pi/4$,

$$x(t + s) - x(t)x(s) = 0 - \frac{1}{2} = -\frac{1}{2}.$$

There is no contradiction: compression of a product need not equal the product of the compressed factors when the hidden subspace carries intermediate amplitude.

The check verifies the differential and integral equations symbolically and the displayed nonzero semigroup defect exactly. The plotted curves are merely a visualization of these closed-form identities.

> **Read this as:** Exact elimination trades a hidden coordinate for memory and initial-data forcing. Replacing that memory by a constant generator is an additional approximation, not part of the exact theorem.

SPECTRAL POSITIVITY IS BASIS INDEPENDENT

Positive transfer is not a stochastic matrix

Main text: Theorem 10.1, p. 18.

For $H = I - X$, the two eigenvalues are zero and two. Its Euclidean transfer is

$$T(\tau) = e^{-\tau H} = \frac{1}{2}\begin{pmatrix} 1+q & 1-q \\ 1-q & 1+q \end{pmatrix}, \qquad q = e^{-2\tau}.$$

For $\tau \geq 0$, the eigenvalues $1, q$ lie in $(0, 1]$, and multiplying two such matrices replaces q, r by qr. This is the semigroup law in spectral coordinates.

T at $e^{-2\tau} = 1/2$

$\frac{3}{4}$	$\frac{1}{4}$
$\frac{1}{4}$	$\frac{3}{4}$

Both have spectrum $\{1, 1/2\}$.

ZTZ in another basis

$\frac{3}{4}$	$-\frac{1}{4}$
$-\frac{1}{4}$	$\frac{3}{4}$

Positive-definite operator: $v^*Tv > 0$ for $v \neq 0$. Not an entrywise test.

At $q = 1/2$, the displayed T happens also to have nonnegative entries and column sums one. Conjugate by the unitary Z. The spectrum and positive-definite quadratic form are preserved, but both off-diagonal entries become $-1/4$. Thus positivity as an operator is not entrywise positivity and does not by itself give a Markov interpretation.

A useful finite positive kernel

For this nonnegative H, vectors u_a and times $t_a \geq 0$ satisfy

$$\sum_{a,b} \langle u_a, e^{-(t_a+t_b)H} u_b \rangle = \left\| \sum_a e^{-t_a H} u_a \right\|^2 \geq 0.$$

The factorization makes positivity transparent. It remains a finite operator identity. A continuum reconstruction theorem requires a compatible field structure, limits, positivity hypotheses, and the other assumptions of the chosen reconstruction framework. None follows just from two positive eigenvalues.

The exact check verifies the matrix semigroup law as a polynomial identity in q, r and verifies the unchanged spectrum after conjugation. It rejects entrywise nonnegativity for the conjugated matrix.

> **Read this as:** Euclidean positivity refers to quadratic forms and spectral values. A stochastic interpretation or a continuum physical reconstruction needs additional structure.

A FOUR-SITE HEAT TRACE IN CLOSED FORM

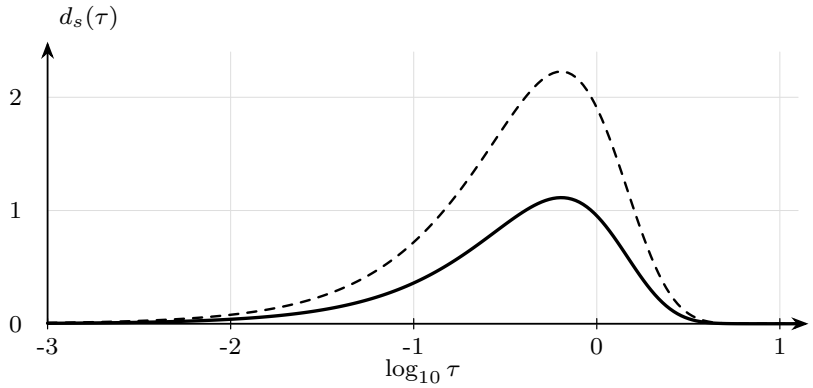
A dimension curve is a scale diagnostic

Main text: Section 11, pp. 21–22.

For the four-cycle Laplacian $L = 2I - S - S^*$, the eigenvalues are $0, 2, 2, 4$. The normalized heat trace and its spectral-dimension diagnostic are therefore

$$K(\tau) = \frac{1 + 2e^{-2\tau} + e^{-4\tau}}{4} = \frac{(1 + e^{-2\tau})^2}{4}, \qquad d_s(\tau) = -2\tau \frac{K'(\tau)}{K(\tau)} = \frac{8\tau}{e^{2\tau} + 1}.$$

No numerical differentiation is needed for this finite example.



solid: four-cycle
dashed: product of two

No broad plateau is promised at four sites.

Finite diagnostic, not a dimension proof.

Both ends are finite-size regimes

As $\tau \to 0$, every eigenmode contributes approximately one and the dimension diagnostic tends to zero. As $\tau \to \infty$, the zero mode dominates, so it again tends to zero. A finite spectrum cannot sustain a nonzero plateau at all scales.

For the Cartesian product of two identical four-cycles, the Laplacian eigenvalues add and the heat traces multiply: $K_2 = K^2$. Consequently $d_{s,2} = 2d_s$ exactly. The second curve is not fitted to the first; the product identity determines it.

At $\tau = \log 2/2$, the first curve equals $4 \log 2/3$. This noninteger value is a scale-dependent spectral quantity, not a claim that the graph has a fractional topological dimension. Four sites are too few to promise a broad geometric plateau. The manuscript's larger-grid experiments are also finite diagnostics, distinguished from its unproved continuum statements.

> **Read this as:** A convincing continuum dimension needs a stable scale window that persists under controlled refinement. One appealing finite curve is not that theorem.

AN EXACT ERROR ENCLOSURE, NOT A FITTED SLOPE

Refine the grid while holding the mode fixed

Main text: Theorem 11.1, p. 21; nonzero-spacing clarification in the formal ledger.

For the circle symbol, the continuum value is k^2 and the grid value is

$$\lambda_{k,h} = \frac{4}{h^2}\sin^2\left(\frac{kh}{2}\right), \qquad h \neq 0.$$

For a fixed low mode, the paper bounds the error by $0 \leq k^2 - \lambda_{k,h} \leq k^4 h^2/12$. The spacing condition matters because the displayed expression divides by h^2.

$h = \frac{1}{4}$	$h = \frac{1}{8}$	$h = \frac{1}{16}$
error (approx.)	error (approx.)	error (approx.)
$\simeq 0.005197495$	$\simeq 0.001301405$	$\simeq 0.000325478$
bound $h^2/12$	bound $h^2/12$	bound $h^2/12$
$\frac{1}{192}$	$\frac{1}{768}$	$\frac{1}{3072}$

Successive error ratios lie strictly between 3.99 and 4.

What is certified in the displayed table?

Take $k = 1$ and $h = 1/4, 1/8, 1/16$. For $x = h/2 \in [0, 1]$, alternating Taylor sums give rational lower and upper bounds for $\sin x$. Squaring these positive endpoints and multiplying by $4/h^2$ yields exact rational enclosures for each error. The displayed decimals only locate those much narrower rational intervals.

Each interval lies strictly inside $(0, h^2/12)$. Dividing successive interval endpoints proves that both error ratios are between 3.99 and 4. The check verifies these inequalities by rational arithmetic, not by trusting floating-point sine values or fitting a line on a logarithmic plot.

Why the low-mode qualification matters

If instead $k = \pi/h$ moves with the grid, the ratio $\lambda_{k,h}/k^2 = 4/\pi^2$ stays constant, not one. The bound's factor k^4 warns that it is not uniform in arbitrary moving high modes. The displayed rational test uses the analytic symbol; it does not assert an integer periodic grid for every chosen spacing.

> **Read this as:** Second-order convergence is a quantified statement with a specified mode regime. Keep that regime fixed when reading an error curve or comparing refinements.

TEST THE CONSTRAINTS BEFORE THE PROJECTION

A zero compressed anomaly can be automatic

Main text: Section 12, p. 23; ConstraintAudit in the finite formal release.

Take the self-adjoint matrices

$$C_1 = \operatorname{diag}(0, 1, -1), \qquad C_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad P = \operatorname{diag}(1, 0, 0).$$

Both constraints annihilate $P\mathbb{C}^3$, and their common kernel is exactly that one-dimensional space. Nevertheless, their full commutator is nonzero.

$[C_1, C_2]$

0	0	0
0	0	2
0	−2	0

$P = \operatorname{diag}(1, 0, 0)$

$C_1 P = C_2 P = 0$

$P[C_1, C_2]P$

0	0	0
0	0	0
0	0	0

The right matrix is zero by the common-kernel choice, not by operator closure.

Why the compressed test cannot decide closure

Since $C_1 P = C_2 P = 0$, each product $C_1 C_2 P$ and $C_2 C_1 P$ vanishes. Therefore $P[C_1, C_2]P = 0$ without any requirement that the unprojected commutator close in the chosen constraint span. In this example the skew off-diagonal commutator is not a linear combination of the two displayed symmetric constraints.

More generally, an anomaly expression made from their commutators and linear combinations of constraints also kills an exact common-kernel projector on the right. This is an important boundary of the diagnostic, recorded by the formal release. A nonzero compressed residual can reveal inconsistent proposed data, but an automatic zero does not construct an anomaly-free constraint algebra.

A substantive closure analysis must inspect the operator relations, their domains, the actual physical subspace, and compatibility under refinement. For infinite systems, these questions also involve domain and limit control that finite matrix arithmetic cannot replace.

The exact example computes both constraint products, the full commutator, and its compression. It deliberately rejects the inference from compressed zero to unprojected zero.

> **Read this as:** Before celebrating a vanishing residual, ask whether the way its projector was defined already forced it to vanish. A tautological diagnostic cannot supply the missing existence or closure theorem.

SEEDS, CHARACTERS, INTERFERENCE, AND DEPHASING

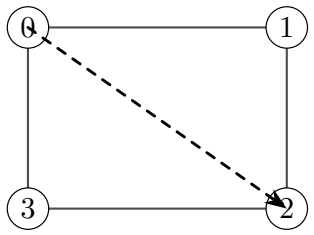
One complete calculation: opposite-site transfer

Main text: Sections 2–5 and 7, especially Theorems 3.1 and 4.1.

Return to the four-site shift S and Laplacian $L = 2I - S - S^*$. The same character projectors have energies $(0, 2, 4, 2)$. At time $t = \pi/2$, their evolution phases are $(1, -1, 1, -1)$, exactly the character eigenvalues of S^2. Hence

$$e^{-\mathrm{i}\pi L/2} = S^2, \qquad |0\rangle \longmapsto |2\rangle.$$

$e^{-\mathrm{i}\pi L/2} = S^2$



start	time $\pi/2$
1	0
0	0
0	1
0	0

Target probability after partial momentum dephasing: $1 - 3\eta/4$.

At halfway time $\pi/4$, the actual amplitude vector is $(-\mathrm{i}, 1, \mathrm{i}, 1)^T/2$, giving uniform site probabilities. The subsequent phases recombine those four amplitudes into a single opposite-site output. Uniform intermediate probabilities do not imply an incoherent intermediate state.

A channel changes the conclusion predictably

Before evolving, mix the seed density matrix with its complete momentum dephasing:

$$\rho_\eta = (1-\eta)|0\rangle\langle 0| + \eta \sum_j P_j |0\rangle\langle 0| P_j = (1-\eta)|0\rangle\langle 0| + \eta I/4, \quad 0 \le \eta \le 1.$$

At transfer time, the opposite-site probability is exactly $1 - 3\eta/4$. Complete momentum dephasing reduces it to $1/4$. This calculation isolates how discarded coherence changes the interference result.

Finally, the shift grammar generates four independent vectors from $|0\rangle$, while the Hamiltonian powers generate only three: its repeated energy links two momentum directions. The same seed can therefore be cyclic for the shift algebra and noncyclic for the Hamiltonian algebra, even though perfect transfer occurs at one time.

> **Read this as:** Perfect transfer is an exact finite interference statement. It neither implies cyclicity for every operator grammar nor survives an unaccounted dephasing operation.

AN EVIDENCE GUIDE FOR THE READER

What these pictures establish

Source release: v0.6.0 finite-completion supplement, pp. 5–7; original manuscript, Section 12.

The appendix uses small objects to make the paper's constructions inspectable. Every matrix and finite example is supplied in editable source, with a separate check record. The plates visualize those calculations; their apparent smoothness or symmetry is not itself proof.

Evidence	What it supports	What it does not supply
Exact matrices	The displayed finite identities, witnesses, and counterexamples.	A classification of arbitrary systems.
Symbolic derivatives	The stated closed-form rotating and Bloch examples.	A proof inferred from numerical differentiation.
Rational enclosures	The three displayed dispersion intervals and ratio inequalities.	A continuum theorem from three sample points.
Saved Lean release	Its 19 registered finite items under their recorded hypotheses.	A fresh kernel run for this appendix.
Numerical replay	The original finite software examples in the tested environment.	Physical experiments or unrestricted convergence.

Which source is authoritative?

The selected formal checkpoint is v0.6.0: 323 recorded theorem declarations in 25 current modules, including supporting lemmas. Its unchanged original manuscript is read with its corrections and completion supplement. The separate v0.5.1 proofread document is retained as an editorial sibling, not merged into those counts. No original proof source is modified by this appendix.

The five research boundaries remain separate: an anomaly-free coherent refinement, uniform effective-operator control, Lorentzian reconstruction, an Einstein infrared limit, and a positive physical massless spin-two sector. A finite projector bundle, positive transfer matrix, or spectral diagnostic does not settle them.

A practical way to use the appendix

Choose one page, reproduce its matrix calculation, and then read the linked theorem with its hypotheses. In particular, retain the two resolvent assumptions on V – 21, the channel-family distinction on V – 20, and the common-kernel caveat on V – 26.

> **Read this as:** Exact finite structure is valuable because its boundary is explicit. The appendix adds understanding and reproducibility, not unproved continuum claims or a new theorem count.